

Practice Questions:

LEVEL I

(Questions Based on Fundamentals)

ABC OF QUADRATIC EQUATION

1. Solve for x

(i) $x^2 - 4x + 3 = 0$

(ii) $4x^2 - 3x - 1 = 0$

(iii) $5x^2 - 15x + 11 = 0$

(iv) $x^2 - 3x + 5 = 0$

(v) $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$

(vi) $4x - 5 \cdot 2^x + 4 = 0$

(vii) $x^{2/3} + x^{1/3} - 2 = 0$

(viii) $7^{1+x} + x^{1/3} - 2 = 0$

(ix) $2x = 42^{x-1}$

(x) $\sqrt{x} + \sqrt{x - \sqrt{1-x}} = 1$.

2. If $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \text{to } \infty}}}$, then find x .

3. If $x = 2 + 2^{2/3} + 2^{1/3}$, then find the value of $x^3 - 6x^2 + 6x$.

4. Solve for x :

$$x^4 - x^3 + 2x^2 - x + 1 = 0$$

5. Solve for x :

$$(x+1)(x+2)(x+3)(x+4) = 120$$

6. If a, b, c are three real numbers, then find the roots of the equation

$$\frac{(x-b)(x-c)}{(a-b)(a-c)}a^2 + \frac{(x-a)(x-c)}{(b-a)(b-c)}b^2 + \frac{(x-b)(x-a)}{(c-b)(c-a)}c^2 - x^2 = 0$$

7. Find the number of real solutions of

$$x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$$

8. Find the number of real roots of the equation

$$(x-1)^2 + (x-2)^2 + (x-3)^2 = 0.$$

9. Find the roots of x , if

$$\frac{x-ab}{a+b} + \frac{x-ac}{a+c} + \frac{x-bc}{b+c} = a+b+c$$

10. Find the roots of x , if $\frac{x-a}{b+c} + \frac{x-b}{c+a} + \frac{x-c}{a+b} = 3$.

11. Solve for x :

$$\sqrt{5x^2 - 6x + 8} - \sqrt{5x^2 - 6x - 7} = 1$$

12. Find the number of roots of x , if $\frac{x-b}{a-b} + \frac{x-a}{b-a} = 1$.
13. Find the values of a , if
 $(a^2 - 3a + 2)x^2 + (|a| - 1)x + (a^2 - 5a + 4) = 0$
 gives more than two roots.

NATURE OF ROOTS

14. Prove that the roots of the equation $x^2 - 2(a+b)x + 2(a^2 + b^2) = 0$ are imaginary and distinct.
15. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ be equal, then prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.
16. For what values of m , the roots of the equation $x^2 - 2(1 + 3m)x + 7(3 + m) = 0$ will be equal?
 The product of the roots of the equation $x^2 - 3\lambda x + 2e^{2\log \lambda} - 1 = 0$ is 7. If the roots be real, then prove that $\lambda = 2$.
18. If the roots of the equation $(b-c)x^2 + (c-a)x + (a-b) = 0$ be equal, then prove that a, b, c are in A.P.
19. If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ has equal roots, prove that a, b, c are in H.P.
 If a, b, c are positive and are in A.P., prove that the roots of the quadratic equation $ax^2 + bx + c = 0$ are real for $\left| \frac{c}{a} - 7 \right| \geq 4\sqrt{3}$.
21. Discuss the nature of the roots of the equation $4ax^2 + 3bx + 2c = 0$, where $a, b, c \in R$ and are connected by the relation $a + b + c = 0$.
22. If the roots of the equation $x^2 - 2cx + ab = 0$ be real and unequal, then prove that the roots of $x^2 - 3(a+b)x + (a^2 + b^2 + 2c^2) = 0$.
 If the roots of the equation $(a-1)(x^2 + x + 1) = (a+1)(x^4 + x^2 + 1)$ are real and distinct, then prove that $a^2 - 4 > 0$.

SUM AND PRODUCT OF THE ROOTS / Transformation

23. If α and β are the roots of the equation $x^2 - 5x + 6 = 0$, then find the value of $\beta^2 + \beta^3, \beta^3 + \beta^4, \alpha - \beta, \alpha^5 + \beta^5, \alpha^7 + \beta^7$.
25. If the product of the roots of the equation $mx^2 - 2x + (2m-1) = 0$ is 3, then find the value of m .
26. If the equation $(k-2)x^2 - (k-4)x - 2 = 0$ has difference of roots as 3, then find the value of k .
27. If α, β and the roots of the equation $ax^2 + bx + c = 0$, then find the value of

$$\frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2}$$

28. If r be the ratio of the roots of the equation $ax^2 + bx + c = 0$, then prove that $\frac{(r+1)^2}{r} = \frac{b^2}{ac}$

29. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but in opposite sign, then prove that $p + q = 2r$, and the product of the roots is $-\frac{1}{2}(p^2 + q^2)$.
30. For the equation $3x^2 + px + 3 = 0, p > 0$, if one of the roots is square of the other, then find p .
31. If one root of the equation $ax^2 + bx + c = 0$ be the square of the other, then prove that $b^3 + ac^2 + a^2c = 3abc$.
 If the roots of the equation $3x^2 + 2(k^2 + 1)x + (k^2 - 3k + 2) = 0$ be of opposite signs, then find k .
32. In a triangle $PQR, \angle R = \pi/2$. If $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation $ax^2 + bx + c = 0, a \neq 0$, then prove that $a + b + c$.
33. Let p and q be the roots of the equation $x^2 - 2x + A = 0$ and let r and s be the roots of the equation $x^2 - 18x + B = 0$. If $p < q < r < s$ are in A.P., then Find A and B .
34. If the roots of the equation $ax^2 + cx + c = 0$ are in the ratio $p : q$, then prove that

$$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{c}{a}} = 0$$

35. If α, β are the roots of the equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$, then find the value of $(\alpha - \gamma)(\alpha - \delta)$.
36. If α, β are the roots of the equation $x^2 + px + 1 = 0$ and γ, δ are the roots of the equation $x^2 + qx + 1 = 0$, then find the value of $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$.
37. If α, β are the roots of $x^2 + px - q = 0$ and γ, δ are the roots of $x^2 + px + r = 0$, prove that $(\alpha - \gamma)(\alpha - \delta) = (\beta - \gamma)(\beta - \delta) = q + r$.
38. If α, β are the roots of $x^2 - x + p = 0$ and γ, δ are the roots of $x^2 - 4x + q = 0$.
 If $\alpha, \beta, \gamma, \delta$ are in G.P., then find the integral values of p and q .
39. If one root of the equation $x^2 - px + q = 0$ be twice the other, show that $2p^2 = 9q$.
40. If the difference of the roots of $x^2 - px + q = 0$ is unity, then prove that
 (a) $p^2 - 4q = 1$
 (b) $p^2 + 4q^2 = (1 + 2q)^2$
41. If α, β are the roots of the equation $x^2 - x - 1 = 0$, then find the value of $\sum \left(\frac{1 - \alpha}{1 + \alpha} \right)$.
42. If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$, then prove that

$$\left(\alpha - \frac{1}{\beta\gamma} \right) \left(\beta - \frac{1}{\gamma\alpha} \right) \left(\gamma - \frac{1}{\alpha\beta} \right) = -\frac{(1+r)^3}{r^2}$$

43. If α, β are the roots of $6x^2 - 6x + 1 = 0$, then prove that

$$\frac{1}{2}(p + q\alpha + r\alpha^2 + s\alpha^3) + \frac{1}{2}(p + q\beta + r\beta^2 + s\beta^3) \text{ is } \frac{p}{1} + \frac{q}{2} + \frac{r}{3} + \frac{s}{4}.$$

44. If α, β are the roots of $ax^2 + 2bx + c = 0$ and $\delta, \delta, \beta + \delta$ are the roots of $Ax^2 + 2Bx + C = 0$, then prove that

$$\frac{b^2 - ac}{B^2 - AC} = \left(\frac{a}{A}\right)^2.$$

45. Let a, b, c, d be real numbers in G.P. If u, v, w satisfy the system of equations $u + 2v + 3w = 6$, $4u + 5v + 6w = 12$, $6u + 9v = 4$, then prove that the roots of the equation $\left(\frac{1}{u} + \frac{1}{v} + \frac{1}{w}\right)x^2 + [(b-c)^2 + (c-a)^2 + (a-b)^2]x + u + v + w = 0$ and $20x^2 + 10(a-d)2x - 9 = 0$ are reciprocals to each other.

FORMATION OF QUADRATIC EQUATION WITH GIVEN ROOTS

46. Find the quadratic equation whose one root is $2 - i\sqrt{3}$.
47. If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then find the equation whose roots are $\alpha^2 - 3\alpha + 7$ and $\beta^2 - 3\beta + 7$.
48. If α, β are roots of the equation $x^2 - 5x + 6 = 0$ then find the equation whose roots are $\alpha + 3$ and $\beta + 3$.
49. If α, β, γ are the roots of the equation $9x^3 - 7x + 6 = 0$, then find the equation whose roots are $3\alpha + 2, 3\beta + 2, 3\gamma + 2$.
50. If α, β are the roots of $2x^2 - 3x - 6 = 0$ then find the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$.
51. If $\alpha \neq \beta$ and $\alpha^2 = 5\alpha - 3, \beta^2 = 5\beta - 3$ form the quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.
52. If the roots of the equation $(x-a)(x-b) = 0$ be c and d , then prove that the roots of the equation $(x-c)(x-d) + k = 0$ are a and b .
53. If α, β are the roots of the equation $(x-a)(x-b) + c = 0$, then find the roots of the equation $(x-\beta)(x-\beta) = c$.
54. α, β are the roots of the equation $\gamma(x^2 - x) + 5 = 0$. If λ_1 and λ_2 be the two values of λ , determined from the equation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$, then prove that the value of $\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} = 254$.
55. If α, β are the roots of $x^2 - (x+1)p - c = 0$ then find the value of

$$\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + c} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + c}$$

COMMON ROOTS OF TWO QUADRATIC EQUATIONS

56. If the equation $x^2 + 2x + 3\lambda = 0, 2x^2 + 3x + 5\lambda = 0$ have a non zero common root, then find λ .
57. If the equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0, a \neq c$ have a negative common root, then find the values of $a - b + c$.
58. If the equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root, then find the value of $a + b$.
59. If the equations $ax^2 + 2cx + b = 0$ and $ax^2 + 2bx + c = 0$ ($b \neq c$) have a common root, then prove that $a + 4b + 4c = 0$.
60. If $x^2 + mx + 1 = 0$ and $(a-b)x^2 + (b-c)x + (c-a) = 0$ have both roots common, then prove that
(i) $m = -2$
(ii) b, a, c are in A.P.
(iii) $2a - b - c = 0$.
61. If the equations $ax^2 + bx + c = 0$ and $x^2 + 2x + 3 = 0$ have common root, then prove that $a : b : c = 1 : 2 : 3$.
62. If the equations $ax^2 + bx + c = 0$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots then prove that $a = b = c$. Find the value of a for which the equation $x^3 + ax + 1 = 0$ and $ax^4 + ax^2 + 1 = 0$ have a common root.

SIGN OF A QUADRATIC EXPRESSION

64. Solve for x :
(i) $x^2 - 2x - 3 < 0$
(ii) $x^2 - 3x + 2 > 0$
65. For what values of k is the inequality $x^2 - (k-3)x - k + 6 > 0$ valid for all real x ?
66. For what values of k , the inequality $(k-2)x^2 + 8x + k + 4 < 0$ satisfies for all real values of x ?
67. For what values of m , the equation $mx^2 - (m+1)x + 2m - 1 = 0$ does not possess any real roots. For what values of p the curves $y = 2px + 1$ and $y = (p-6)x^2 - 2$ do not intersect?
69. For what values of k , the curve $y = x^2 + kx + 4$ touches the x -axis?
70. Find the integral values of k for which the equation $(k-12)x^2 + 2(k-12)x + 2 = 0$ possess no real roots.
- 71 Find the value of ' b ' $x^2 + bx + 1 > 0$.
- 72 Find a , if $x^2 + 2ax + 10 - 3a > 0, \forall x \in \mathbb{R}$.
- 73 Find λ , if $x^2 - 2(4\lambda - 1)x + (15\lambda^2 - 2\lambda - 7) > 0, \forall x \in \mathbb{R}$.

RANGE OF A QUADRATIC POLYNOMIAL

74. Find the maximum and minimum values of
(i) $f(x) = x^2 + 2x + 4$
(ii) $f(x) = x^2 + 4x + 4$
(iii) $f(x) = x^2 - 5x + 4$
(iv) $f(x) = -x^2 + x - 4$
(v) $f(x) = -x^2 + 6x - 9$
(vi) $f(x) = -x^2 + 6x - 8$
75. Find the ranges of
(i) $f(x) = x^2 + x + 1$
(ii) $f(x) = x^2 + 3x + 2$

iii) $f(x) = -x^2 + 3x - 10$

76. Let $P(x) = ax^2 + bx + 8$ is a quadratic polynomial. If the minimum value of $P(x)$ is 6 at $x = 2$ then find the value of a and b .
77. Find the range of $f(x) = 2x^2 - 3x + 2$ in $[0, 2]$
78. Find the range of $f(x) = -x^2 + 6x - 1$ in $[0, 4]$

MAXIMUM AND MINIMUM VALUES OF A RATIONAL EXPRESSION

79. If $x \in \mathbb{R}$, find the least and greatest value of the expression $\frac{x^2 - 6x + 1}{x^2 + 6x + 1}$.

80. If $y = \frac{x^2 - 2x + 4}{x^2 + 2x + 4}$, $x \in \mathbb{R}$, then find the max and min value of y .

81. If $y = \frac{x^2 - x + 1}{x^2 + x + 1}$, $x \in \mathbb{R}$, then find y .

82. Find the greatest and least value of $y = \frac{x^2 + x + 1}{x^2 - x + 1}$.

83. If x is real, find the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$.

84. If $x \in \mathbb{R}$, then prove that, $\frac{x^2 + 2x + a}{x^2 + 4x + 3a}$ can take all real values if $a \in [0, 1]$.

85. If $y = \tan x \cot 3x$, $x \in \mathbb{R}$, then find y .

86. Find the range of the expression

$$f(\theta) = \frac{\tan^2 \theta - 2 \tan \theta - 8}{\tan^2 \theta - 4 \tan \theta - 5}$$

87. Find the range of the expression

$$f(\theta) = \frac{(\cot^2 \theta + 5)(\cot^2 \theta + 10)}{(\cot^2 \theta + 1)}$$

88. Find the range of values of a , such that

$$f(x) = \frac{ax^2 + 2(a+1)x + 9a - 4}{x^2 - 8x + 32} \text{ is always negative.}$$

Show that the function $z = 2x^2 + 2xy + y^2 - 2x + 2y + 2$ is not smaller than -3 .

90. Find the minimum value of

$$f(x) = \frac{\left(x + \frac{1}{x}\right)^6 - \left(x^6 + \frac{1}{x^6}\right) - 2}{\left(x + \frac{1}{x}\right)^3 + \left(x^3 + \frac{1}{x^3}\right)} \text{ for } x > 0$$

91. Let x be a positive real. Find the maximum possible value of the expression

$$y = \frac{x^2 + 2 - \sqrt{x^4 + 4}}{x}$$

RESOLUTION INTO TWO LINEAR FACTORS

92. Find the values of m for which the expression $2x^2 + mxy + 3y^2 - 5y - 2$ may be resolved into two linear factors.
93. Find the value of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ may resolve into two linear factors.
94. If the expression $ax^2 + by^2 + cz^2 + 2ayz + 2bzx + 2cxy$ be resolved into two rational factors, then prove that, $a^3 + b^3 + c^3 = 3abc$.
95. If x, y and z are real and distinct, then prove that $f(x, y) = x^2 + 4y^2 + 9z^2 - 6yz - 3zx - 2xy$ is always non-negative.
96. If x is real and $4y^2 + 4xy + x + 6 = 0$, then find the complete set of values of x for which y is real. Find the greatest and least values of x and y satisfying the relation $x^2 + y^2 = 6x - 8y$.

LOCATION OF ROOTS

98. Find the values of m for which both roots of equation $x^2 - mx + 1 = 0$ are less than unity.
99. For what real values of m both roots of the equation $x^2 - 6mx + 9m^2 - 2m + 2 = 0$ exceed 3?
100. Find all values of p so that 6 lies between roots of the equation $x^2 + 2(p-3)x + 9 = 0$
101. Find the values of m for which exactly one root of the equation $x^2 - 2mx + m^2 - 1 = 0$ lies in the interval $(-2, 4)$.
102. Find all values of a for which the equation $4x^2 - 2x + a = 0$ has two roots lie in the interval $(-1, 1)$
103. Find the values of a for which one root of the equation $(a-5)x^2 - 2ax + a - 4 = 0$ is smaller than 1 and the other greater than 2.
104. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3 then find a .
105. If both the roots of the equation $x^2 - 12kx + k^2 + k - 5 = 0$ are less than 5, then prove that
106. If both the roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, find a
107. If the equation $ax^2 + bx + c = 0$ ($a > 0$) has two roots α and β such that $\alpha < -2$ and $\beta > 2$ then
- $b^2 - 4ac > 0$
 - $4a + 2|b| + c < 0$
 - $a + |b| + c = 0$
 - $c < 0$
108. Find the value of ' λ ' for which $2x^2 - 2(2\lambda + 1)x + \lambda(\lambda + 1) = 10$ may have one root less than λ and other root greater than λ .
109. Find the value of ' a ' for which the equation $2x^2 - 2(2a + 1)x + a(a - 1) = 0$ has roots α and β such that $\alpha < a < \beta$.

LOCATION OF ROOTS IN INEQUALITIES

110. Find all values of m for which the inequality $mx^2 - 4x + 3m + 1 > 0$ is satisfied for all +ve values of x .

111. Find all values of b for which $x^2 - x + b - 3 < 0$ for atleast one $-ve$ x .
112. Find all values of k for which the inequality $k \cdot 4^x + (k - 1) \cdot 2^{x+2} + (k - 1) > 1$ is satisfied for all x in R .
113. Find all possible parameters ' a ' for which $f(x) = (a^2 + a - 2)x^2 - (a + 5)x - 2$ is non $+ve$ for every $x \in [0, 1]$.

SOME SPECIAL TYPE EQUATIONS

114. $x^4 - 5x^2 + 4 = 0$
115. $2x^4 + 2x^2 + 3 = 0$.
116. $(x - 1)^4 + (x - 5)^4 = 82$
117. $2x^4 - x^3 - 11x^2 - x + 2 = 0$
118. $(x - 1)(x - 2)(x - 3)(x - 4) = 8$.
119. $x^2 + \frac{x^2}{(x + 1)^2} = 3$.

TRANSFORMATION EQUATIONS

120. Solve : $3x^3 - 22x^2 + 48x - 72 = 0$ when the roots are in H.P.
121. Solve : $6x^3 - 11x^2 + 6x - 1 = 0$, when the roots are in H.P.
122. If α, β, γ be the roots of $x^3 + px^2 + qx + r = 0$, then find a cubic equation whose roots are $\alpha(\beta + \gamma), \beta(\gamma + \alpha), \gamma(\alpha + \beta)$.
123. If α, β, γ be the roots of $x^3 - px^2 + r = 0$, then find a cubic equation whose roots are $\frac{\beta + \gamma}{\alpha}, \frac{\gamma + \alpha}{\beta}, \frac{\alpha + \beta}{\gamma}$.
124. If α, β, γ be the roots of $x^3 + x^2 - 4x + 7 = 0$, then find a cubic equation whose roots are $\beta + \gamma, \gamma + \alpha, \alpha + \beta$.
125. If α, β, γ be the roots of $x^3 + x + 2 = 0$, then find a cubic equation whose roots are $(\alpha - \beta)^2, (\beta - \gamma)^2, (\gamma - \alpha)^2$.

DESCARTES RULES OF SIGN

26. Find the number of $+ve$ and $-ve$ roots of $x^5 - x^4 + x^3 + 8x^2 + 2x - 2 = 0$.
127. Find the number and position of the real roots of $x^4 - 41x^2 + 40x + 126 = 0$.
128. Find the number and position of the real roots of $x^4 - 14x^2 + 16x + 9 = 0$.
129. Find the nature of the roots of $3x^4 + 12x^2 + 5x - 4 = 0$.
130. Find the least possible number of imaginary roots of $x^9 - x^5 + x^4 + x^2 + 1 = 0$.
131. Show that $x^7 - 3x^4 + 2x^3 - 1 = 0$ has atleast four imaginary roots..

IN-EQUATION

Type I: ABC of In-equation

Solve for x :

132. $x^2 - 4x + 3 > 0$.
133. $x^2 + x - 2 \geq 0$.
134. $x^2 + 2x - 8 < 0$.
135. $x^2 - 7x + 10 \leq 0$.
136. $x^2 - 3x + 2 > 0$.

137. $x^2 - 5x + 4 \leq 0$.
138. $x^2 - 7x + 12 \leq 0$.
139. $x^2 - 8x + 12 < 0$.
140. $x^3 - 6x^2 + 11x - 6 < 0$.
141. $x^3 - 4x \leq 0$.
142. $x^3 - 6x^2 + 12x - 8 \geq 0$.

Type II: Co-efficient of the highest power of x is positive.

Q. Solve for x :

143. $-x^2 + 5x - 4 > 0$.
144. $-x^2 + 7x - 6 \leq 0$.
145. $-x^2 + 3x - 2 > 0$.
146. $-x^2 + 5x - 6 < 0$.
147. $-x^2 + 10x - 21 \leq 0$.
148. $-x^3 + x \leq 0$.
149. $-x^3 + 4x \geq 0$.
150. $(x - 2)(1 - x^2) \geq 0$.
151. $-x^3 + x^2 - 2x \geq 0$.

Type III: Transpose is allowed but cross-multiplication is not allowed.

Q. Solve for x :

152. $\frac{1}{x} > 2$
153. $\frac{x - 2}{4 - x} \geq 0$
154. $\frac{(x - 1)(x - 3)}{(x + 1)(x - 5)} \geq 0$
155. $\frac{1}{x - 2} \geq \frac{1}{x + 1}$
156. $\frac{1}{x} > 1$
157. $\frac{2}{x} \geq 1$
158. $\frac{1}{x} > \frac{1}{x - 1}$
159. $\frac{1}{x + 2} \leq \frac{1}{x - 3}$
160. $\frac{1}{x} - 1 \geq 0$
161. $\frac{(x - 1)(x - 2)}{(x - 3)(x - 4)} \geq 0$
162. $\frac{(x - 1)(x - 2)}{(x - 2)(x - 4)} \leq 0$
163. $\frac{1}{x + 1} - 1 \geq 2 - \frac{1}{x - 2}$

Type IV: If $f(x)$ contains an even power factors, we shall not represent on the number line, we neglect them.

Q. Solve for x :

164. $x^2(x - 2)^4(x - 3) \geq 0$
165. $(x - 1)^4(x - 3)^6(x - 2)^5(x - 4)^{10} > 0$

$$166. \frac{(x-1)^{2013}(x-2)^{2014}}{(x-3)^{2016}(x-5)^{2010}} > 0$$

$$167. x^2(x-1) > 0$$

$$168. x^2(x-1)^{10}(x+2)^{11} > 0$$

$$169. \frac{(x-1)^4(x-2)^{12}}{(x-3)^{17}(x-5)^{2012}} > 0$$

$$170. \frac{(x+1)^{201}(x-4)^{2012}}{(x+3)^{2012}(x-2)^{2013}} > 0$$

$$171. \frac{1}{x^2} - 1 \geq 0$$

$$172. x^3 - 3x + 2 \geq 0$$

$$173. x^3 + 7x^2 - 36 \geq 0$$

Type V: If $f(x) = ax^2 + bx + c$, where $a > 0$ and $D < 0$, then we discard it

Solve for x :

$$174. x^2 + x + 2 > 0$$

$$175. x^2 - x + 3 < 0$$

$$176. (x+2)(x^4 + x^2 + 1) > 0$$

$$177. x^4 - 4 \leq 0$$

$$178. (x^2 + 4x + 1)(x^2 + 1) \geq 0$$

$$179. (x-2)(x^2 + x + 2) > 0$$

$$180. (x-3)(-x^2 + x + 1) > 0$$

$$181. x^3 + 4x \geq 0$$

$$182. x^4 - 9 \geq 0$$

$$183. x^3 - 5x + 4 \geq 0$$

$$184. x^3 - 3x^2 + 3x - 9 \geq 0$$

$$185. x^3 - 6x^2 + 12x - 9 \geq 0$$

$$186. 2x^2 + 5x > 12$$

$$187. 4x^2 + 4x + 1 \leq 0$$

Type VI: Common values of x in

$$f(x) \geq 0 \text{ and } g(x) \geq 0$$

or

$$f(x) \geq 0 \text{ and } g(x) \leq 0$$

Q. Solve for x :

$$188. x^2 - 3x + 2 > 0, x^2 + 2x - 8 < 0$$

$$189. x^2 - 9 \leq 0, x^2 - 1 \geq 0$$

$$190. x^3 - 9x \geq 0 \text{ and } x^3 + 4x \leq 0$$

$$191. x^2 - 4x + 3 \geq 0 \text{ and } x^2 - 4 \leq 0$$

$$192. x^2 - 9 \geq 0 \text{ and } x^2 - 4 \leq 0$$

$$193. x^2 - 6x + 8 \geq 0 \text{ and } x^2 - 3x + 2 \geq 0$$

$$194. x^2 - 5x + 6 \geq 0 \text{ and } x^2 - 10x + 24 \leq 0$$

$$195. x^2 - x \geq 0 \text{ and } x^2 - 12x + 27 \leq 0$$

$$196. x^3 + x \geq 0 \text{ and } x^2 - 9 \leq 0$$

$$197. x^3 - 9x \geq 0 \text{ and } x^3 + 4x \leq 0$$

EQUATION CONTAINING ABSOLUTE VALUES

$$198. \text{Solve for } x: |x|^2 - 3|x| + 2 = 0$$

$$199. \text{Find the sum of the roots of } x^2 - 4|x| + 3 = 0$$

Solve for x :

$$200. |3x - 1| = |x + 5|$$

$$201. |2x - 5| = x - 3$$

$$202. |x^2 - x - 6| = x + 2$$

$$203. 2|x - 2| + 3|x - 4| = 3$$

$$204. |x| + |x - 2| = 4$$

$$205. |x - 1| + |x - 3| = 2$$

$$206. |x^2 - 1| + |x^2 - 4| = 3$$

$$207. \sqrt{x + 2\sqrt{x - 1}} + \sqrt{x - 2\sqrt{x - 1}} = 2$$

$$208. \left| \frac{x}{x-1} \right| + |x| = \frac{x^2}{|x-1|}$$

$$209. 2^{|x+1|} - 2^x = |2^x - 1| + 1$$

$$210. |x^2 + x - 20| = -(x^2 + x - 20)$$

$$211. \left| \left(\frac{x^2 - 6x + 8}{x^2 - 4x + 3} \right) \right| = - \left(\frac{x^2 - 6x + 8}{x^2 - 4x + 3} \right)$$

INEQUALITIES WITH THE ABSOLUTE VALUE

Solve for x :

$$212. |x - 1| < 3$$

$$213. |x - 4| \leq 3$$

$$214. |x + 2| > 5$$

$$215. |x - 2| \geq 3$$

$$216. \left| \frac{4x - 2}{3} \right| \leq 2$$

$$217. x^2 - |3x - 2| > 0$$

$$218. |x| + |x - 2| > 3$$

$$219. |x + 2| + |x| + |x - 2| > 6.$$

$$220. \left| \frac{2}{x + 3} \right| \leq 1$$

$$221. |3x + 2| > |2x - 1|$$

IRRATIONAL EQUATIONS

Solve for x :

$$222. \sqrt{2x + 7} + \sqrt{x + 4} = 0$$

$$223. \sqrt{x - 4} = -5$$

$$224. \sqrt{x - 6} - \sqrt{8 - x} = 2$$

$$225. \sqrt{-2 - x} = \sqrt[3]{x - 7}$$

$$226. \sqrt{x} + \sqrt{x + 16} = 3$$

$$227. 7\sqrt{x} + 8\sqrt{-x} + \frac{15}{x^3} = 98.$$

$$228. \sqrt{x - 2} + \sqrt{4 - x} = \sqrt{6 - x}$$

$$229. \sqrt{2x - 4} - \sqrt{x + 5} = 1$$

$$230. \sqrt{3x + 4} + \sqrt{x - 4} = 2\sqrt{x}$$

$$231. \sqrt{x - 1} + \sqrt{2x + 6} = 6$$

IRRATIONAL INEQUATIONS

 Solve for x :

232. $\sqrt[5]{\frac{3}{x+1} + \frac{7}{x+2}} < \sqrt[5]{\frac{6}{x-1}}$

233. $\sqrt[3]{\frac{x-2}{x-1}} < \sqrt[3]{\frac{1}{x-1}}$

234. $\sqrt{3x-2} < \sqrt{x+4}$

235. $\sqrt{4x-3} < \sqrt{2x+5}$

236. $\sqrt[3]{(3x-5)} < (x-1)$

237. $\sqrt[3]{3x-2} < x$

238. $\sqrt{(x+14)} < (x+2)$

239. $\sqrt{2x-2} < x-1$

240. $\sqrt[3]{5x-4} > x$

241. $\sqrt[3]{x^3 - 7x^2} > (-36)^{1/3}$

242. $\sqrt{-x^2 + 4x - 3} > (6 - 2x)$

243. $\sqrt{x-2} + \sqrt{x-1} > 2$

244. $\sqrt{x-6} - \sqrt{10-x} \geq 1$

EXPONENTIAL EQUATIONS

 Solve for x :

245. $5^{x^2+3x+2} = 1$

246. $3^{x^2+5|x|+6} = 1$

247. $5^{2x} - 24 \cdot 5^x - 25 = 0$

248. $64 \cdot 9^x - 84 \cdot 12^x + 27 \cdot 16^x = 0$

249. $15 \cdot 2^{x+1} + 15 \cdot 2^{2-x} = 135$

250. $3^{x-4} + 5^{x-4} = 34$

251. $1 + 3^{x/2} = 2^x$

252. $3^x + 4^x + 5^x = 6^x$

253. $1^x + 6^x + 8^x = 9^x$

254. $16^{\sin^2 x} + 16^{\cos^2 x} = 10$

255. $2^x + 2^{x-1} + 2^{x-2} = 5^x + 5^{x-1} + 5^{x-2}$

256. $(5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10$

257. $(15 + 4\sqrt{14})^x + (15 - 4\sqrt{14})^x = 30$

258. $(5^x + 5^{-x}) = \log_{10} 25, x \in \mathbb{R}$

259. $\sin x = x^2 + x + 1$

260. Find x , for every real x , $p^x + p^{-x} = \log_q q^2 - 1$ where $p, q \in \mathbb{R}$.

261. Find the number of solutions of x , $n^{-|x|}m - |x| = 1$, where $m, n > 1$ and $n > m$.

262. Find the number of solution of x for which $2x + x^2 = 1$.

EXPONENTIAL IN-EQUATIONS

 Solve for x :

263. $2^x > 1$

264. $\left(\frac{1}{3}\right)^x > 1$

265. $5^{x^2+3x+3} > 5$

266. $\left(\frac{1}{2}\right)^{x^2-5x+8} > \left(\frac{1}{4}\right)$

267. $4^x + 2^{x+1} - 8 \leq 0$

268. $|x|^{x^2-x-2} \geq 1, -1 < x < 1$

269. $\left(\frac{1}{3}\right)^{\sqrt{x+4}} > \left(\frac{1}{3}\right)^{\sqrt{x^2+3x+4}}$

270. $\frac{2^{1-x} - 2^x + 1}{2^x - 1} \leq 0$

271. $2^{x+1} - 2^{x+2} - 2^{x+3} > 5^{x+1} - 5^{x+2}$

272. $3^{x^2-4} < 2^{x^2-x}$

Hints and Solutions:

LEVEL I

2. Let $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$ to ∞

$$x = \sqrt{2 + x}$$

$$x^2 = 2 + x$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = 2, -1$$

since square always provide us non negative values, so
 $x = 2$

3. Given $x = 2 + 2^{2/3} + 2^{1/3}$

$$x - 2 = 2^{2/3} + 2^{1/3}$$

$$(x - 2)^3 = (2^{2/3} + 2^{1/3})^3$$

$$x^3 - 6x^2 + 12x - 8$$

$$= (2^{2/3})^3 + (2^{1/3})^3 + 3 \cdot 2^{2/3} \cdot 2^{1/3} (2^{2/3} + 2^{1/3})$$

$$= 4 + 2 + 3 \cdot 2(x - 2)$$

$$= 6 + 6x - 12$$

$$= 6x - 6$$

$$x^3 - 6x^2 + 6x = 8 - 6 = 3$$

4. Divide by x^2 , we get

$$\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right) + 2 = 0$$

$$\left(x + \frac{1}{x}\right)^2 - 2 - \left(x + \frac{1}{x}\right) + 2 = 0$$

$$\left(x + \frac{1}{x}\right)^2 - \left(x + \frac{1}{x}\right) = 0$$

Put $\left(x + \frac{1}{x}\right) = t$

Then $t^2 - t = 0$

$$t = 0 \text{ and } t = 1$$

$$x + \frac{1}{x} = 0 \text{ and } x + \frac{1}{x} = 1$$

$$x^2 + 1 = 0 \text{ and } x^2 - x + 1 = 0$$

$$x = \pm i \text{ and } x = \frac{1 \pm i\sqrt{3}}{2}$$

Hence, the solutions are

$$\left\{ \pm i, \frac{1 \pm i\sqrt{3}}{2} \right\}$$

5. $(x + 1)(x + 2)(x + 3)(x + 4) = 120$

$$\{(x + 1)(x + 4)\} \{(x + 2)(x + 3)\} = 120$$

$$\{(x^2 + 5x + 4)\} \{(x^2 + 5x + 6)\} = 120$$

Put $x^2 + 5x = a$

$$(a + 4)(a + 6) = 120$$

$$a^2 + 10a - 96 = 0$$

$$(a + 16)(a - 6) = 0$$

$$a = 6, -16$$

$$x^2 + 5x = 6, -16$$

$$x^2 + 5x - 6 = 0, x^2 + 5x + 16 = 0$$

$$(x + 6)(x - 1) = 0, x^2 + 5x + 16 = 0$$

$$x = 1, -6, \frac{-5 \pm i\sqrt{39}}{2}$$

Hence, the solutions are

$$\left\{ 1, -6, \frac{-5 \pm i\sqrt{39}}{2} \right\}$$

6. It is true for all real values of x .

So it is an identity in x

Thus the number of real solution is infinite.

7. $x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$

No real values of x satisfying the given equation

Thus, $x = \phi$

8. $(x - 1)^2 + (x - 1)^2 + (x - 3)^2 = 0$

$$3x^2 - 12x + 14 = 0$$

Clearly, $D < 0$

Number of real solutions is zero

9. $\frac{x - ab}{a + b} + \frac{x - bc}{b + c} + \frac{x - ca}{c + a} = a + b + c$

$$\frac{x-ab}{a+b} - c + \frac{x-bc}{b+c} - a + \frac{x-ca}{c+a} - b = 0$$

$$\frac{x-ab-ac-bc}{a+b} + \frac{x-ab-ac-bc}{b+c} + \frac{x-ab-ac-bc}{c+a} = 0$$

$$(x-ab-ac-bc) \left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} \right) = 0$$

$$(x-ab-ac-bc) = 0$$

$$x = (ab+bc+ca)$$

$$10. \frac{x-a}{b+c} + \frac{x-b}{c+a} + \frac{x-c}{a+b} = 3$$

$$\frac{x-a}{b+c} - 1 + \frac{x-b}{c+a} - 1 + \frac{x-c}{a+b} - 1 = 0$$

$$\frac{x-a-b-c}{b+c} + \frac{x-b-c-a}{c+a} + \frac{x-c-a-b}{a+b} = 0$$

$$(x-a-b-c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) = 0$$

$$(x-a-b-c) = 0$$

$$x = (a+b+c)$$

$$11. \text{ Put } u = \sqrt{5x^2 - 6x + 8}, v = \sqrt{5x^2 - 6x - 7}$$

$$\text{Then } u - v = 1$$

$$u^2 - v^2 = 15$$

and then solve it

$$12. \text{ Given equation is } \frac{x-b}{a-b} + \frac{x-a}{b-a} = 1$$

Clearly, it is true for all real values of x .

So It is an identity in x

Thus, the number of real roots = infinite

$$13. \text{ Since the given equation is an identity in } x, \text{ so}$$

$$a^2 - 3a + 2 = 0, |a| - 1 = 0, a^2 - 5a + 4 = 0$$

$$(a-1)(a-2) = 0, |a| = 1, (a-1)(a-4) = 0$$

$$a = 1, 2; a = \pm 1; a = 1, 4$$

Hence, the value of a is 1. i.e. $a = 1$

$$14. \text{ To prove } D < 0$$

$$15. \text{ Do yourself}$$

$$16. \text{ Apply } D = 0$$

$$17. \text{ It is given that, product of the roots} = 7$$

$$2e^{2\log \lambda} - 1 = 7$$

$$2e^{\log \lambda^2} = 8$$

$$e^{\log \lambda^2} = 4$$

$$\lambda^2 = 4$$

$$\lambda = \pm 2$$

$\lambda = 2$, since the logarithm of a negative number is not defined.

$$18. S = \text{Sum of the co-efficients}$$

$$= b - c + c - a + a - b = 0$$

So, 1 is a root

Now, product of the roots = $1 \cdot 1 = 1$

$$\frac{a-b}{b-c} = 1$$

$$a-b = b-c$$

$$a+c = 2b$$

a, b and c are in A.P

$$19. S = \text{Sum of the co-efficients}$$

$$= a(b-c) + b(c-a) + c(a-b)$$

$$= ab - ac + bc - ba + ca - ab = 0$$

Thus, 1 is a root

Now, product of the roots = $1 \cdot 1 = 1$

$$\frac{c(a-b)}{a(b-c)} = 1$$

$$ac - bc = ab - ac$$

$$2ac = ab + bc$$

$$b(a+c) = 2ac$$

$$b = \frac{2ac}{(a+c)}$$

Thus, a, b and c are in H.P

$$20. \text{ Given } a, b, c \text{ are in A.P}$$

$$a+c = 2b$$

Also, it is given that, $D \geq 0$

$$b^2 - 4ac \geq 0$$

$$\left(\frac{a+c}{2} \right)^2 - 4ac \geq 0$$

$$a^2 + c^2 - 14ac \geq 0$$

$$\left(\frac{c}{a} \right)^2 - 14 \left(\frac{c}{a} \right) + 1 \geq 0$$

$$\left(\frac{c}{a} - 7 \right)^2 \geq (4\sqrt{3})^2$$

$$\left| \left(\frac{c}{a} - 7 \right) \right| \geq (4\sqrt{3})$$

$$21. \text{ Do yourself}$$

$$22. \text{ Given roots of } x^2 - 2cx + ab = 0 \text{ are real}$$

$$\text{So, } 4c^2 - 4ab \geq 0$$

$$c^2 - ab \geq 0$$

$$\text{Now, } D = 4(a+b)^2 - 4(a^2 + b^2 + 2c^2)$$

$$= 4\{(4(a+b)^2 - 4(a^2 + b^2 + 2c^2))\}$$

$$= 4\{a^2 + b^2 + 2ab - (a^2 + b^2 + c^2)\}$$

$$= 4(2ab - 2c^2)$$

$$= 8(ab - c^2) < 0$$

Thus, the roots are imaginary

$$23. \text{ Given equation is}$$

$$(a-1)(x^2+x+1)^2 = (a+1)(x^4+x^2+1)$$

$$(a-1)(x^2+x+1)^2 = (a+1)(x^2-x+1)(x^2+x+1)$$

$$(a-1)(x^2+x+1) = (a+1)(x^2-x+1)$$

$$-2x^2 + 2ax - 2 = 0$$

$$x^2 - ax + 1 = 0$$

Since roots are real and distinct,

so, $D > 0$

$$a^2 - 4 > 0$$

24. Do yourself
25. Do yourself
26. Do yourself
27. Do yourself

28. Given $\frac{\alpha}{\beta} = r$

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

$$\beta(1+r) = -\frac{b}{a}$$

$$\beta = -\frac{b}{a(1+r)}$$

Also, $\alpha\beta = \frac{c}{a}$

$$r\beta^2 = \frac{c}{a}$$

$$r \frac{b^2}{a^2(1+r)^2} = \frac{c}{a}$$

$$\frac{b^2 r}{(1+r)^2} = ac$$

$$\frac{b^2}{ac} = \frac{(1+r)^2}{r}$$

Hence, the result.

29. Given equation is

$$\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$$

$$\frac{x+q+x+p}{(x+p)(x+q)} = \frac{1}{r}$$

$$(x+p)(x+q) = (2x+p+q)r$$

$$x^2 + (p+q)x + pq = (2x+p+q)r$$

$$x^2 + (p+q-2r)x + pq - r(p+q) = 0$$

Sum of the roots = 0

$$(p+q-2r) = 0$$

$$p+q = 2r$$

Now, product of the roots

$$= pq - r(p+q)$$

$$= pq - \frac{1}{2}(p+q)^2$$

$$= -\frac{1}{2}(p^2 + q^2)$$

30. Let the roots be α, α^2

Now, product of the roots = 1

$$\alpha \cdot \alpha^2 = 1$$

$$\alpha^3 = 1$$

$\alpha = 1, \omega, \omega^2$, where ω is the cube root of unity.

Let $\alpha = \omega, \omega^2$

Now, sum of the roots = $-\frac{p}{3}$

$$(\omega + \omega^2) = -\frac{p}{3}$$

$$-1 = -\frac{p}{3}$$

$$p = 3$$

Hence, the value of p is 3

31. Let the roots be α, α^2

$$\alpha + \alpha^2 = -\frac{b}{a}, \alpha \cdot \alpha^2 = \frac{c}{a}$$

$$\alpha + \alpha^2 = -\frac{b}{a}, \alpha^3 = \frac{c}{a}$$

Now, $\alpha + \alpha^2 = -\frac{b}{a}$

$$(\alpha + \alpha^2)^3 = \left(-\frac{b}{a}\right)^3$$

$$\alpha^3 + \alpha^6 + 3\alpha^3(\alpha + \alpha^2) = \left(-\frac{b}{a}\right)^3$$

$$\frac{c}{a} + \frac{c^2}{a^2} + 3 \cdot \frac{c}{a} \left(-\frac{b}{a}\right) = -\frac{b^3}{a^3}$$

$$b^3 + ac^2 + a^2c = 3abc$$

Hence, the result

32. Product of the roots < 0

$$(k^2 - 3k + 2) < 0$$

$$(k-1)(k-2) < 0$$

$$1 < k < 2$$

33. It is given that, $\angle P + \angle Q = \frac{\pi}{2}$

$$\frac{P}{2} + \frac{Q}{2} = \frac{\pi}{4}$$

$$\tan\left(\frac{P}{2} + \frac{Q}{2}\right) = \tan\left(\frac{\pi}{4}\right) = 1$$

$$\frac{\tan\left(\frac{P}{2}\right) + \tan\left(\frac{Q}{2}\right)}{1 - \tan\left(\frac{P}{2}\right) \cdot \tan\left(\frac{Q}{2}\right)} = 1$$

$$\frac{-\frac{b}{a}}{1 - \frac{c}{a}} = 1$$

$$1 - \frac{c}{a}$$

$$-b = a - c$$

$$a + b = c$$

Hence, the result.

33. Let $p = a - 3d, q = a - d, r = a + d, s = a + 3d$

Now, $p + q = 2, pq = A; r + s = 18, rs = B$

Now, $p + q + r + s = 20$

$$4a = 20$$

$$a = 5$$

Also, $p = q = 2$

$$2a - 4d = 2$$

$$10 - 4d = 2$$

$$4d = 8$$

$$d = 2$$

$$p = a - 3d = 5 - 6 = -1$$

$$q = a - d = 5 - 2 = 3$$

Therefore,

$$r = a + d = 5 + 2 = 7$$

$$s = a + 3d = 5 + 6 = 11$$

$$\text{Thus, } A = pq = -3$$

$$B = rs = 77$$

$$34. \text{ Given } \frac{\alpha}{\beta} = \sqrt{\frac{p}{q}}$$

$$\text{Here, } \alpha + \beta = -\frac{c}{a}, \alpha\beta = \frac{c}{a}$$

$$\begin{aligned} \text{Now, } \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{c}{a}} \\ &= \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{c}{a}} \\ &= \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + \sqrt{\frac{c}{a}} \\ &= \frac{-\frac{c}{a}}{\sqrt{\frac{c}{a}}} + \sqrt{\frac{c}{a}} \\ &= -\sqrt{\frac{c}{a}} + \sqrt{\frac{c}{a}} \\ &= 0 \end{aligned}$$

Hence, the result.

$$35. \quad x^2 + px + q = 0 \quad x^2 + px - r = 0$$

$$\alpha, \beta \quad \gamma, \delta$$

$$\alpha + \beta = -p, \alpha\beta = q \quad \gamma + \delta = -p, \gamma\delta = -r$$

$$\begin{aligned} \text{Now, } (\alpha - \gamma)(\alpha - \delta) \\ &= \alpha^2 - (\gamma + \delta)\alpha + \gamma\delta \\ &= \alpha^2 + p\alpha - r \\ &= -q - r \\ &= -(q + r) \end{aligned}$$

$$36. \quad x^2 + px + 1 = 0 \quad x^2 + qx + 1 = 0$$

$$\alpha, \beta \quad \gamma, \delta$$

$$\alpha + \beta = -p, \alpha\beta = 1 \quad \gamma + \delta = -q, \gamma\delta = 1$$

$$\begin{aligned} \text{Now, } (\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) \\ &= (\gamma - \alpha)(\gamma - \beta)(\delta + \alpha)(\delta + \beta) \\ &= \{\gamma^2 - (\alpha + \beta)\gamma + \alpha\beta\} \{\delta^2 + (\alpha + \beta)\delta + \alpha\beta\} \\ &= \{\gamma^2 + p\gamma + 1\} \{\delta^2 - p\delta + 1\} \\ &= (p\gamma - q\gamma)(-q\delta - p\delta) \\ &= -(p - q)(p + q)\gamma\delta \\ &= -(p - q)(p + q) \\ &= -(p^2 - q^2) \\ &= (q^2 - p^2) \end{aligned}$$

37. Similiar to Q.36

$$38. \quad x^2 - x + p = 0 \quad x^2 - 4x + q = 0$$

$$\alpha, \beta \quad \gamma, \delta$$

$$\alpha + \beta = 1, \alpha\beta = p \quad \gamma + \delta = 4, \gamma\delta = q$$

It is given that, $\alpha, \beta, \gamma, \delta$ are in G.P

$$\alpha, \alpha r, \alpha r^2, \alpha r^3 \in \text{G.P}$$

$$\text{Now, } \alpha + \beta = 1$$

$$\alpha + \alpha r = 1$$

$$\alpha(1 + r) = 1$$

$$\alpha = \frac{1}{(1 + r)}$$

$$\text{Also, } \gamma + \delta = 4$$

$$\alpha r^2 + \alpha r^3 = 4$$

$$\alpha r^2(1 + r) = 4$$

$$r^2 = 4$$

$$r = \pm 2$$

$$\text{when } r = 2, \alpha = \frac{1}{3} \quad \text{when } r = -2, \alpha = -1$$

$$\beta = \alpha r = \frac{2}{3} \quad \beta = \alpha r = 2$$

$$\gamma = \alpha r^2 = \frac{4}{3} \quad \gamma = \alpha r^2 = -4$$

$$\delta = \alpha r^3 = \frac{8}{3} \quad \delta = \alpha r^3 = 8$$

$$\text{Therefore, } p = \alpha\beta = -2$$

$$q = \gamma\delta = -32$$

39. Let the roots be α, β

It is given that $\alpha = 2\beta$

$$\text{Now, } \alpha + \beta = p, \alpha\beta = q$$

$$2\beta + \beta = p, \alpha\beta = q$$

$$3\beta = p, 2\beta^2 = q$$

$$\beta = \frac{p}{3}, 2\beta^2 = q$$

$$2\beta^2 = q$$

$$2\left(\frac{p}{3}\right)^2 = q$$

$$2p^2 = 9q$$

40. It is given that, $|\alpha - \beta| = 1$

$$|\alpha - \beta|^2 = 1$$

$$(\alpha + \beta)^2 - 4\alpha\beta = 1$$

$$p^2 - 4q = 1$$

$$\text{Also, } p^2 = 4q + 1$$

$$p^2 + 4q^2 = 4q^2 + 4q + 1 = (2q + 1)^2$$

Hence, the result.

41. Here, $\alpha + \beta = -1, \alpha\beta = -1$

$$\begin{aligned} \text{Now, } \sum \left(\frac{1 - \alpha}{1 + \alpha} \right) &= \left(\frac{1 - \alpha}{1 + \alpha} \right) + \left(\frac{1 - \beta}{1 + \beta} \right) \\ &= \frac{2 - 2\alpha\beta}{1 + (\alpha + \beta) + \alpha\beta} \\ &= \frac{2 + 2}{1 - 1 - 1} = -4 \end{aligned}$$

42. Here,

$$\alpha + \beta + \gamma = -p, \alpha\beta + \alpha\gamma + \beta\gamma = q, \alpha\beta\gamma = -r$$

$$\begin{aligned} \text{Now, } & \left(\alpha - \frac{1}{\beta\gamma}\right)\left(\beta - \frac{1}{\alpha\gamma}\right)\left(\gamma - \frac{1}{\alpha\beta}\right) \\ &= \left(\alpha - \frac{\alpha}{\alpha\beta\gamma}\right)\left(\beta - \frac{\beta}{\alpha\beta\gamma}\right)\left(\gamma - \frac{\gamma}{\alpha\beta\gamma}\right) \\ &= \alpha\beta\gamma\left(1 - \frac{1}{\alpha\beta\gamma}\right)^3 \\ &= -r\left(1 + \frac{1}{r}\right)^3 \\ &= -\frac{(1+r)^3}{r^2} \end{aligned}$$

43. Here, $\alpha + \beta = 1, \alpha\beta = \frac{1}{6}$

Now,

$$\begin{aligned} & \frac{1}{2}(p + q\alpha + r\alpha^2 + s\alpha^3) + \frac{1}{2}(p + q\beta + r\beta^2 + s\beta^3) \\ &= \frac{1}{2}\{(p + p) + q(\alpha + \beta) + r(\alpha^2 + \beta^2) + s(\alpha^3 + \beta^3)\} \\ &= \frac{1}{2}\left(2p + q + \frac{2}{3}r + \frac{s}{2}\right) \\ &= \frac{p}{1} + \frac{q}{2} + \frac{r}{3} + \frac{s}{4} \end{aligned}$$

44. Here, $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

$$\text{and } (\alpha + \delta) + (\beta + \delta) = -\frac{2B}{A}, (\alpha + \delta)(\beta + \delta) = \frac{C}{A}$$

$$\text{Now, } \alpha - \beta = (\alpha + \delta) - (\beta + \delta)$$

$$(\alpha - \beta)^2 = \{(\alpha + \delta) - (\beta + \delta)\}^2$$

$$(\alpha + \beta)^2 - 4\alpha\beta = \{(\alpha + \delta) + (\beta + \delta)\}^2 - 4(\alpha + \delta)(\beta + \delta)$$

$$\frac{4b^2}{a^2} - \frac{4c}{a} = \frac{4B^2}{A^2} - \frac{4C}{A}$$

$$\frac{b^2 - ac}{a^2} = \frac{B^2 - AC}{A^2}$$

$$\frac{b^2 - ac}{B^2 - AC} = \left(\frac{a}{A}\right)^2$$

Hence, the result.

45. See the solution of Q.57 (IIT-JEE-1999)

$$46. x^2 - Sx + P = 0$$

$$x^2 - 4x + 7 = 0$$

$$47. \alpha^2 - 3\alpha + 5 = 0 \Rightarrow \alpha^2 - 3\alpha + 7 = 2$$

$$\beta^2 - 3\beta + 5 = 0 \Rightarrow \beta^2 - 3\beta + 7 = 2$$

Hence, the required equation is

$$x^2 - Sx + P = 0$$

$$x^2 - 4x + 4 = 0$$

$$48. x^2 - 5x + 6 = 0 \text{ gives } x = 2, x = 3$$

$$\text{Let } \alpha = 2, \beta = 3$$

Hence, the required equation is

$$x^2 - Sx + P = 0$$

$$x^2 - 11x + 30 = 0$$

$$49. \text{ Let } y = 3\alpha + 2$$

$$\alpha = \frac{y-2}{3}$$

$$\text{Now, } 9\alpha^3 - 7\alpha + 2 = 0$$

$$9\left(\frac{y-2}{3}\right)^3 - 7\left(\frac{y-2}{3}\right) + 2 = 0$$

$$9(y-2)^3 - 7(y-2) + 18 = 0$$

$$9y^3 - 54y^2 + 101y - 50 = 0$$

Hence, the required equation is

$$9x^3 - 54x^2 + 101x - 50 = 0$$

50. Do yourself

51. Do yourself

52. We have

$$(x-a)(x-b) - k = (x-c)(x-d)$$

$$(x-c)(x-d) + k = (x-a)(x-b)$$

Hence, the roots are a and b.

53. We have

$$(x-a)(x-b) + c = (x-\alpha)(x-\beta)$$

$$(x-a)(x-b) - c = (x-a)(x-b)$$

Hence, the roots are a and b

54. The given equation reduces to

$$\lambda x^2 + (1-\lambda)x + 5 = 0$$

Let its roots are α and β

$$\alpha + \beta = \frac{\lambda-1}{\lambda}, \alpha\beta = \frac{5}{\lambda}$$

$$\text{Now, } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{4}{5}$$

$$\frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{4}{5}$$

$$\frac{(\lambda-1)^2 - \frac{10}{\lambda}}{\frac{5}{\lambda}} = \frac{4}{5}$$

$$\frac{(\lambda-1)^2 - 10\lambda}{5\lambda} = \frac{4}{5}$$

$$\lambda^2 - 16\lambda + 1 = 0$$

Let its roots are λ_1 and λ_2

$$\text{Thus, } \lambda_1 + \lambda_2 = 16, \lambda_1\lambda_2 = 1$$

$$\text{Now, } \frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}$$

$$= \frac{(\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2}{\lambda_1\lambda_2}$$

$$= \frac{256 - 2}{1}$$

$$= 254$$

55. The given equation is

$$x^2 - (x+1)p - c = 0$$

$$x^2 - px - (p+c) = 0$$

$$\alpha + \beta = p, \alpha\beta = -(p+c)$$

 Now, $(1+\alpha)(1+\beta)$

$$= 1 + (\alpha + \beta) + \alpha\beta$$

$$= 1 + p - p - c$$

$$= 1 - c$$

$$\text{Now, } \frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + c} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + c}$$

$$= \frac{(\alpha+1)^2}{(\alpha+1)^2 - (1-c)} + \frac{(\beta+1)^2}{(\beta+1)^2 - (1-c)}$$

$$= \frac{(\alpha+1)^2}{(\alpha+1)^2 - (1+\alpha)(1+\beta)} + \frac{(\beta+1)^2}{(\beta+1)^2 - (1+\alpha)(1+\beta)}$$

$$= \frac{(\alpha+1)}{(\alpha+1) - (1+\beta)} + \frac{(\beta+1)}{(\beta+1) - (1+\alpha)}$$

$$= \frac{(\alpha+1)}{(\alpha-\beta)} + \frac{(\beta+1)}{(\beta-\alpha)}$$

$$= \frac{(\alpha+1)}{(\alpha-\beta)} - \frac{(\beta+1)}{(\alpha-\beta)}$$

$$= \frac{(\alpha-\beta)}{(\alpha-\beta)}$$

$$= 1$$

56. Do yourself

57. $ax^2 + bx + c = 0$... (i)

$cx^2 + bx + a = 0$... (ii)

 (i) - (ii), we get, $(a-c)x^2 = (a-c)$

$$x^2 = 1$$

$$x = \pm 1$$

$$x = -1$$

Hence, the value of

$$a - b + c = 0$$

58. $x^2 + ax + b = 0$... (i)

$x^2 + bx + a = 0$... (ii)

(i) - (ii), we get,

$$(a-b)x = (a-b)$$

$$x = 1$$

Hence, the value of

$$a + b = -1$$

59. $ax^2 + 2cx + b = 0$... (i)

$ax^2 + 2bx + c = 0$... (ii)

(i) - (ii), we get,

$$2(c-b)x = (c-b)$$

$$x = \frac{1}{2}$$

 Put the value of $x = \frac{1}{2}$ in (i), we get,

$$\frac{a}{4} + c + b = 0$$

$$a + 4b + 4c = 0$$

60. Let S = sum of the co-efficients

$$= a - b + b - c + c - a$$

$$= 0$$

Thus, 1 is a root

 Now, $1 + m + 1 = 0$

$$m = -2$$

Also, $\frac{c-a}{a-b} = 1$

$$c - a = a - b$$

$$b + c = 2a$$

$$2a - b - c = 0$$

61. $D = 4 - 12 = -8 < 0$

Clearly, both roots are common

Thus, $\frac{a}{1} = \frac{b}{2} = \frac{c}{3}$

$$a : b : c = 1 : 2 : 3$$

62. Do yourself

63. Do yourself

64. (i) $x^2 - 2x - 3 < 0$

$$(x-3)(x+1) < 0$$

$$-1 < x < 3$$

(ii) $x^2 - 3x + 2 > 0$

$$(x-1)(x-2) > 0$$

$$x < 1 \text{ and } x > 2$$

65. The given inequation is

$$x^2 - (k-3)x - (k-6) > 0$$

 Clearly, $D < 0$

$$(k-3)^2 + 4(k-6) < 0$$

$$k^2 - 2k - 15 < 0$$

$$(k-5)(k+3) < 0$$

$$-3 < k < 5$$

 66. Clearly, $k - 2 < 0$

$$k < 2$$

... (i)

 Also, $D < 0$

$$64 - 4(k-2)(k+4) < 0$$

$$(k-2)(k+4) - 16 > 0$$

$$k^2 + 2k - 24 > 0$$

$$(k+6)(k-4) > 0$$

$$k < -6 \text{ and } k > 4$$

... (ii)

From (i) and (ii), we get,

$$k < -6$$

 67. Clearly, $D < 0$

$$(m+1)^2 - 4m(2m-1) < 0$$

$$7m^2 - 6m - 1 < 0$$

$$(m-1)(7m+1) < 0$$

$$m \in \left(-\frac{1}{7}, 1\right)$$

 68. Clearly, $2px + 1 = (p-6)x^2 - 2$

$$(p-6)x^2 - 2px - 3 = 0$$

 Obviously, $D < 0$

$$4p^2 + 12(p-6) < 0$$

$$p^2 + 3(p-6) < 0$$

$$p^2 + 3p - 18 < 0$$

$$(p+6)(p-3) < 0$$

$$-6 < p < 3$$

69. Clearly, $D = 0$

$$k^2 - 16 = 0$$

$$k = \pm 4$$

70. Clearly, $D < 0$

$$4(k-12)^2 - 8(k-12) < 0$$

$$(k-12)^2 - 2(k-12) < 0$$

$$(k-12)(k-14) < 0$$

$$12 < k < 14$$

71. The given in equation is

$$x^2 + bx + 1 > 0$$

It is true only when $b^2 - 4 < 0$

$$(b+2)(b-2) < 0$$

$$-2 < b < 2$$

72. The given in equation is

$$x^2 + 2ax + 10 - 3a > 0$$

Clearly, $D < 0$

$$4a^2 - 40 + 12a < 0$$

$$a^2 + 3a - 10 < 0$$

$$(a+5)(a-2) < 0$$

$$-5 < a < 2$$

73. Clearly, $D < 0$

$$4(4\lambda-1)^2 - 4(15\lambda^2 - 2\lambda - 7) < 0$$

$$\lambda^2 - 6\lambda + 18 < 0$$

$$(\lambda-2)(\lambda-4) < 0$$

$$2 < \lambda < 4$$

74. (i) $x^2 + 2x + 4 = (x+1)^2 + 3 \geq 3$

$$\text{Min } V = 3$$

(ii) $x^2 + 4x + 4 = (x+2)^2 \geq 0$

$$\text{Min } V = 0$$

75. (i) $x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$

$$\text{Min } V = 3/4$$

(ii) $x^2 + 3x + 2 = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4} \geq -\frac{1}{4}$

$$\text{Min } V = -1/4$$

76. We have $f(x) = ax^2 + bx + 8$

$$= a \left\{ \left(x + \frac{b}{2a}\right)^2 + \frac{8}{a} - \frac{b^2}{4a^2} \right\}$$

$$= a \left\{ \left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 32a}{4a^2} \right\}$$

$$\text{Clearly, } \frac{b}{2a} = -2$$

$$b = -4a$$

$$\frac{32a - b^2}{4a} = 6$$

$$\frac{32a - 16a^2}{4a} = 6$$

$$8a - 4a^2 = 6a$$

$$4a^2 = 2a$$

$$a = \frac{1}{2}$$

$$b = -2$$

77. We have $f(x) = 2x^2 - 3x + 2$

$$= 2 \left(x^2 - \frac{3}{2}x + 1 \right)$$

$$= 2 \left(x - \frac{3}{2} \right)^2 + \frac{7}{8}$$

Min value is $7/8$

$$\text{Also, } f(0) = 2, f(2) = 4$$

$$\text{Thus, Range} = \left[\frac{7}{8}, 4 \right]$$

78. We have $f(x) = -x^2 + 6x - 1$

$$= -(x-3)^2 + 8$$

Max value = 8

$$\text{Also, } f(0) = -1, f(4) = 7$$

$$\text{Thus, Range} = [-1, 8]$$

79. Let $y = \frac{x^2 - 6x + 1}{x^2 + 6x + 1}$

$$y = \frac{\left(x + \frac{1}{x}\right) - 6}{\left(x + \frac{1}{x}\right) + 6}$$

$$\text{Let } g(x) = \left(x + \frac{1}{x}\right)$$

$$g'(x) = 1 - \frac{1}{x^2}$$

$$g'(x) = 0 \text{ gives } 1 - \frac{1}{x^2} = 0$$

$$x = \pm 1$$

$$\text{Min } V = -2 \text{ at } x = -1$$

$$\text{Max } V = -1/2 \text{ at } x = 1$$

80. Similar to 79

81. Similar to 79

82. Similar to 79

83. Let $y = \frac{3x^2 + 9x + 17}{3x^2 + 9x + 7} = 1 + \frac{10}{3x^2 + 9x + 7}$

$$y \leq 1 + 10.4 = 41$$

Hence, the maximum value of y is 41.

where, minimum value of $3x^2 + 9x + 7$ is $\frac{1}{4}$

Thus, the maximum value of $\frac{1}{3x^2 + 9x + 7}$ is 4

84. Let $y = \frac{x^2 + 2x + a}{x^2 + 4x + 3a}$

$$(y-1)x^2 + 2(2y-1)x + a(3y-1) = 0$$

Clearly, $D \geq 0$

$$(2y-1)^2 - a(y-1)(3y-1) \geq 0$$

$$(4-3a)y^2 + 4(a-1)y + (5-a) \geq 0$$

Thus, $(4-3a) > 0$, $16(a-1)^2 - 4(1-a)(4-3a) \geq 0$
 $(4-3a) > 0$, $16(a-1)^2 - 4(1-a)(4-3a) < 0$
 $(4-3a) > 0$, $4(a-1)^2 - (1-a)(4-3a) < 0$
 $4(-3a) > 0$, $a(a-1) < 0$

$$a < \frac{3}{4}, 0 < a < 1$$

Hence, the value of a is $(0, 1)$

85. We have

$$y = \frac{\tan x}{\tan 3x} = \frac{1-3t^2}{3-t^2}, t = \tan x$$

$$\frac{3y+1}{y-3} = t^2 > 0$$

$$\frac{3y+1}{y-3} > 0$$

$$y < -\frac{1}{3}, y > 3$$

86. Let $y = \frac{\tan^2 \theta - 2 \tan \theta - 8}{\tan^2 \theta - 4 \tan \theta - 5}$

$$y = \frac{t^2 - 2t - 8}{t^2 - 4t - 5}$$

$$(y-1)t^2 + 2(1-2y)t + (8-5y) = 0$$

Clearly, $D \geq 0$

$$4(1-2y)^2 - 4(y-1)(8-5y) \geq 0$$

$$(1-2y)^2 - (y-1)(8-5y) \geq 0$$

$$9y^2 - 17y + 9 \geq 0$$

Clearly, $D < 0$

Thus it is true for all y

Hence, the range = $\mathbb{R}$

87. See the solution of Q-2 (Level-IV)

88. See the solution of Q-28 (Level-IV)

89. Do yourself

90. See the solution of Q-26 (Level-IV)

91. See the solution of Q-27 (Level-IV)

92. Here, $a = 2$, $b = 3$, $c = -2$, $h = \frac{m}{2}$, $g = 0$, $f = \frac{5}{2}$

We have,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$-12 + 2 \times 0 - 2 \times \frac{25}{4} - 0 + 2 \times \frac{m^2}{4} = 0$$

$$\frac{m^2}{2} - \frac{25}{2} - 12 = 0$$

$$\frac{m^2}{2} = \frac{49}{2}$$

$$m^2 = 49$$

$$m = \pm 7$$

93. Do yourself

94. Do yourself

95. We have $f(x, y, z)$

$$= x^2 + 4y^2 + 9z^2 - 6yz - zx - 2xy$$

$$= x^2 + (2y)^2 + (3z)^2 - (2y \cdot 3z) - (x \cdot z) - (x \cdot 2y)$$

$$= \frac{1}{2} [(x-2y)^2 + (2y-3z)^2 + (3z-x)^2]$$

$$\geq 0$$

96. We have $4y^2 + 4xy + x + 6 = 0$

$$4y^2 + 4xy + (x+6) = 0$$

Clearly, $D \geq 0$

$$16x^2 - 16(x+6) \geq 0$$

$$x^2 - (x+6) \geq 0$$

$$(x-3)(x+2) \geq 0$$

$$x \leq -2, x \geq 3$$

97. See the solution of Q-5 of Level - III

98. Let $f(x) = x^2 - mx + 1$

Apply

(i) $D \geq 0$

$$\Rightarrow m^2 - 4 \geq 0$$

$$\Rightarrow (m+2)(m-2) \geq 0$$

$$\Rightarrow m \leq -2 \text{ and } m \geq 2$$

(ii) $af(1) > 0$

$$\Rightarrow (1-m+1) < 0$$

$$\Rightarrow m < 2$$

(iii) $\alpha + \beta < 2 \Rightarrow m < 2$

From (i), (ii) and (iii), we get

$$m < -2$$

99. Let $f(x) = x^2 - 6mx + 9m^2 - 2m + 2$

Apply

(i) $D \geq 0$

$$36m^2 - 36m^2 + 8m - 8 \geq 0$$

$$m \geq 1$$

(ii) $a \cdot f(3) > 0$

$$1 \cdot f(3) > 0$$

$$f(3) > 0$$

$$9 - 18m + 9m^2 - 2m + 2 > 0$$

$$9m^2 - 20m + 11 > 0$$

$$(m-1)(9m-11) > 0$$

$$m < 1 \text{ and } m > \frac{11}{9}$$

(iii) $\alpha + \beta > 6$

$$6m > 6$$

$$m > 1$$

From (i), (ii) and (iii), we get, $m \in \left(\frac{11}{9}, \infty\right)$

100 Let $f(x) = x^2 + 2(p-3)x + 9$

Apply

(i) $D > 0$

$$4(p-3)^2 - 36 > 0$$

$$(p-3)^2 - 9 > 0$$

$$p(p-6) > 0$$

$$p < 0 \text{ and } p > 6$$

(ii) $af(6) < 0$

$$f(6) < 0$$

$$36 + 12(p-3) + 9 < 0$$

$$12p + 9 < 0$$

$$4p + 3 < 0$$

$$p < -\frac{3}{4}$$

From (i) and (ii), we get,

$$p \in \left(-\infty, -\frac{3}{4}\right)$$

101. Let $f(x) = x^2 - 2mx + m^2 - 1$

Apply

(i) $D \geq 0$

$$4m^2 - 4(m^2 - 1) \geq 0$$

$$4 > 0$$

It is true all values of m

(ii) $f(-2)f(4) < 0$

$$(m^2 + 4m + 3)(m^2 - 8m + 15) < 0$$

$$(m + 1)(m + 3)(m - 3)(m - 5) < 0$$

$$m \in (-3, -1) \cup (3, 5)$$

From (i) and (ii), we get

$$m \in (-3, -1) \cup (3, 5)$$

102. Let $f(x) = 4x^2 - 2x + a$

Apply

(i) $D \geq 0$

(ii) $af(-1) > 0$

(iii) $af(1) > 0$

(iv) $\alpha < 0 < \beta$

103. Let $f(x) = (a - 5)x^2 - 2ax + (a - 4)$

Apply

(i) $D > 0$

(ii) $af(1) < 0$

(iii) $af(2) < 0$

104. Let $f(x) = x^2 - 2x + a^2 + a - 3$

Apply

(i) $D \geq 0$

(ii) $af(3) > 0$

(iii) $\alpha + \beta < 6$

105. Let $f(x) = x^2 - 12kx + k^2 + k - 5$

Apply

(i) $D \geq 0$

(ii) $af(5) > 0$

(iii) $\alpha + \beta < 10$

106. Let $f(x) = x^2 - 6ax + 9a^2 - 2a + 2$

Apply

(i) $D \geq 0$

(ii) $af(3) > 0$

(iii) $\alpha + \beta > 6$

107. Let $f(x) = ax^2 + bx + c$

Apply

(i) $D > 0$

(ii) $af(-2) < 0$

(iii) $af(2) < 0$

108. Similar to 107

109. Let $f(x) = 2x^2 - 2(2a + 1)x + a(a - 1)$

Apply

(i) $D > 0$

(ii) $2f(a) < 0$

110. We have

$$m > 0 \text{ and } D < 0$$

$$16 - 4m(3m + 1) < 0$$

$$3m^2 + m - 4 > 0$$

$$(3m + 4)(m - 1) > 0$$

$$m < -\frac{4}{3} \text{ and } m > 1$$

Hence, the solution is $m \in (1, \infty)$

112. The given equation is

$$k \cdot 4^x + (k - 1)2^{x+2} + (k - 1) > 0$$

$$k \cdot (2^x)^2 + 4(k - 1)2^x + (k - 1) > 0$$

$$k \cdot (t)^2 + 4(k - 1)t + (k - 1) > 0, t = 2^x$$

Let $f(t) = k \cdot (t)^2 + 4(k - 1)t + (k - 1)$

Apply

(i) $D < 0$

$$4(k - 1)^2 - k(k - 1) < 0$$

$$3k^2 - 7k + 4 < 0$$

$$3k^2 - 3k - 4k + 4 < 0$$

$$3k(k - 1) - 4(k - 1) < 0$$

$$(3k - 4)(k - 1) < 0$$

$$1 < k < \frac{4}{3}$$

(ii) $k > 0, f(0) \geq 0$

$$k > 0, k - 1 \geq 0$$

$$k > 0, k \geq 1$$

$$k \geq 1$$

(iii) $\alpha + \beta < 0$

$$\frac{k - 1}{k} > 0$$

$$k < 0 \text{ and } k > 1$$

From (i), (ii) and (iii), we get,

$$1 < k < \frac{4}{3}$$

114. Put $x^2 = t$

The given equation reduces to

$$t^2 - 5t + 4 = 0$$

$$(t - 4)(t - 1) = 0$$

$$t = 1, 4$$

$$x^2 = 1, 4$$

$$x = \pm 1, \pm 2$$

115. Put $x^2 = t$ and then try to solve it

116. Put $y = \frac{(x - 1) + (x - 5)}{2} = x - 3$

$$x = y + 3$$

The given equation reduces to

$$(y + 2)^4 + (y - 2)^4 = 82$$

$$y^4 + 24y^2 - 25 = 0$$

$$(y^2 + 25)(y^2 - 1) = 0$$

$$y = \pm 5i, \pm 1$$

$$x - 3 = \pm 5i, \pm 1$$

$$x = 3 \pm 5i, 3 \pm 1$$

$$x = 3 \pm 5i, 4, 2$$

117. The given equation is

$$2x^4 - x^3 - 11x^2 - x + 2 = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right) - 11 = 0$$

Put $x + \frac{1}{x} = t$

Then the given equation reduces to

$$2(t^2 - 2) - t - 11 = 0$$

$$2t^2 - t - 15 = 0$$

$$(t - 3)(2t + 5) = 0$$

$$t = 3, -\frac{5}{2}$$

$$x + \frac{1}{x} = 3, -\frac{5}{2}$$

$$x^2 - 3x + 1 = 0, 2x^2 + 5x + 2 = 0$$

$$x = \frac{3 \pm \sqrt{5}}{2}, x = -2, -\frac{1}{2}$$

Hence, the solution set is

$$\left\{-2, -\frac{1}{2}, \frac{3 \pm \sqrt{5}}{2}\right\}$$

118. The given equation is

$$(x - 1)(x - 1)(x - 1)(x - 1) = 8$$

$$\{(x - 1)(x - 4)\} \{(x - 2)(x - 3)\} = 8$$

$$\{(x^2 - 5x + 4)\} \{x^2 - 5x + 6\} = 8$$

$$(a + 4)(a + 6) = 8, x^2 - 5x = a$$

$$(a + 4)(a + 6) = 8$$

$$a^2 + 10a + 16 = 0$$

$$(a + 2)(a + 8) = 0$$

$$a = -2, -8$$

$$x^2 - 5x = -2, -8$$

$$x^2 - 5x + 2 = 0, x^2 - 5x + 8 = 0$$

$$x = \frac{5 \pm \sqrt{17}}{2}, \frac{5 \pm i\sqrt{7}}{2}$$

119. The given equation is

$$x^2 + \frac{x^2}{(x+1)^2} = 3$$

$$x^4 + 2x^3 - x^2 - 6x - 3 = 0$$

Let $f(x) = x^4 + 2x^3 - x^2 - 6x - 3$

$$f(x): ++--$$

$$f(-x): +---$$

In $f(x)$, there are one change, so $f(x)$ has one +ve root and in $f(-x)$, there are 3 changes, so $f(-x)$ has 3 negative roots

Thus, the number of real roots = 1 + 3 = 4

120. Replace x by $\frac{1}{x}$ we get

$$32x^3 - 48x^2 + 22x - 3 = 0$$

Let its roots are $\alpha - \beta, \alpha, \alpha + \beta$

$$\text{Sum of the roots} = \frac{48}{32} = \frac{3}{2}$$

$$3\alpha = \frac{3}{2}$$

$$\alpha = \frac{1}{2}$$

$$\text{Also, } \alpha(\alpha + \beta)(\alpha - \beta) = \frac{3}{32}$$

$$\alpha(\alpha^2 - \beta^2) = \frac{3}{32}$$

$$\frac{1}{2}\left(\frac{1}{4} - \beta^2\right) = \frac{3}{32}$$

$$\beta = \pm \frac{1}{4}$$

Hence, the roots are

$$\frac{1}{2} - \frac{1}{4}, \frac{1}{2}, \frac{1}{2} + \frac{1}{4} \text{ or } \frac{1}{2} + \frac{1}{4}, \frac{1}{2}, \frac{1}{2} - \frac{1}{4}$$

$$\frac{1}{4}, \frac{1}{2}, \frac{3}{4} \text{ or } \frac{3}{4}, \frac{1}{2}, \frac{1}{4}$$

Therefore, the roots of the main equation are

$$4, 2, \frac{4}{3} \text{ or } \frac{4}{3}, 2, 4$$

121. The given equation is

$$6 - 11x + 6x^2 - x^3 = 0$$

Replace x by $1/x$, we get

$$x^3 - 6x^2 + 11x - 6 = 0$$

$$(x - 1)(x - 2)(x - 3) = 0$$

$$x = 1, 2, 3$$

Hence, the solutions are $1, \frac{1}{2}, \frac{1}{3}$

122. Let $y = \alpha\beta + \alpha\gamma + \beta\gamma - \beta\gamma$

$$y = q - \beta\gamma$$

$$y = q - \frac{\alpha\beta\gamma}{\alpha} = q + \frac{r}{\alpha}$$

$$\alpha = \frac{r}{y - q}$$

Since α is a root of the given equation, so

$$\alpha^3 + p\alpha^2 + q\alpha + r = 0$$

$$\left(\frac{r}{y - q}\right)^3 + p\left(\frac{r}{y - q}\right)^2 + q\left(\frac{r}{y - q}\right) + r = 0$$

Hence, the required equation is

$$r^3 + pr(x - q) + q(x - q)^2 + r(x - q)^3 = 0$$

123. Let $y = \frac{\beta + \gamma}{\alpha} = \frac{\alpha + \beta + \gamma - \alpha}{\alpha} = \frac{p}{\alpha} - 1$

$$\alpha = \frac{p}{y + 1}$$

Since α is a root of the given equation, so

$$\gamma^3, p\alpha^2 + r = 0$$

$$\left(\frac{p}{y+1}\right)^3 - p\left(\frac{p}{y+1}\right)^2 + r = 0$$

Hence, the required equation is

$$p^3 - q^3(x-1) + (x+1)^3 = 0$$

$$(x+1)^3 - p^3(x+1) = p^3 = 0$$

$$124. \text{ Let } y = \beta + \gamma = \alpha + \beta + \gamma - \alpha = -1 - \alpha$$

$$\alpha = -(y+1)$$

Since α is a root of the given equation, so

$$\gamma^3 + \gamma^2 + 4\alpha + 7 = 0$$

$$-(y+1)^3 + (y+1)^2 - 4(y+1) + 7 = 0$$

$$(y+1)^3 - (y+1)^2 + 4(y+1) - 7 = 0$$

Hence, the equation is

$$(x+1)^3 - (x+1)^2 + 4(x+1) - 7 = 0$$

$$125. \text{ Let } y = (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= (\alpha + \beta + \gamma - \gamma)^2 - 4\alpha\beta$$

$$= (0 - \gamma)^2 - 4\alpha\beta$$

$$= \gamma^2 - 4\alpha\beta$$

$$= \gamma^2 + \frac{8}{\gamma}$$

$$= \frac{\gamma^3 + 8}{\gamma}$$

$$= \frac{-\gamma - 2 + 8}{\gamma}$$

$$= \frac{-\gamma + 6}{\gamma}$$

$$\gamma = \frac{6}{y+1}$$

Since γ is a root of the given equation, so $\gamma^3 + \gamma + 2 = 0$

$$\left(\frac{6}{y+1}\right)^3 + \left(\frac{6}{y+1}\right) + 2 = 0$$

$$(y+1)^3 + 3(y+1)^2 + 108 = 0$$

Hence, the required equation is

$$(x+1)^3 + 3(x+1)^2 + 108 = 0$$

$$126. \text{ Let } f(x) = x^5 - x^4 + x^3 + 8x^2 + 2x - 2$$

$$f(x): + - + + -$$

$$f(-x): - - - + -$$

There are 3 changes in $f(x)$ and 2 changes in $f(-x)$, so

$f(x)$ has 3 +ve root and 2 negative roots.

Thus, the number of real roots = $3 + 2 = 5$

$$127. \text{ Let } f(x) = x^4 - 41x^2 + 40x + 126$$

$$f(x): + - + +$$

$$f(-x): + - - +$$

There are 2 changes in $f(x)$ and 2 changes in $f(-x)$. So

$f(x)$ has 2 +ve roots and 2 negative roots

Hence, the number of real roots = $2 + 2 = 4$

$$128. f(x) = x^4 - 14x^2 + 16x + 9$$

$$f(x): + - + +$$

$$f(-x): + - - +$$

There are 2 changes in $f(x)$ and 2 changes in $f(-x)$. So

$f(x)$ has 2 +ve roots and 2 negative roots

Hence, the number of real roots = $2 + 2 = 4$

$$129. \text{ Let } f(x) = 3x^4 + 12x^2 + 5x - 4$$

$$f(x): + + + -$$

$$f(-x): + + - -$$

There are 1 change in $f(x)$ and also 1 change in $f(-x)$

So $f(x)$ has 1 +ve root and 1 -ve root

Thus, the number of real roots = 2

$$130. \text{ Let } f(x) = x^9 - x^5 + x^4 + x^2 + 1$$

$$f(x): + - + + +$$

$$f(-x): - + + + +$$

There are 2 changes in $f(x)$ and 1 change in $f(-x)$. So

$f(x)$ has 2 +ve real roots and 1 -ve real root

Thus, the number of real roots = $2 + 1 = 3$

Hence, the number of imaginary roots = $9 - 3 = 6$

$$131. \text{ Let } f(x) = x^7 - 3x^4 + 2x^3 - 1$$

$$f(x): + - + -$$

$$f(-x): - - - -$$

There are 3 changes in $f(x)$ and no change in $f(-x)$

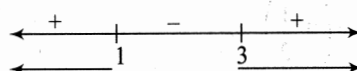
So $f(x)$ has 3 +ve real roots and 0 -ve root

Hence, the number of imaginary roots = $7 - 3 = 4$.

132. Given in-equation is

$$x^2 - 4x + 3 > 0.$$

$$\Rightarrow (x-1)(x-3) > 0$$



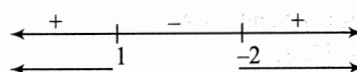
$$\Rightarrow x < 1 \text{ and } x > 3$$

$$\Rightarrow x \in (-\infty, 1) \cup (3, \infty)$$

133. Given in-equation is

$$x^2 + x - 2 \geq 0$$

$$\Rightarrow (x+2)(x-1) \geq 0$$



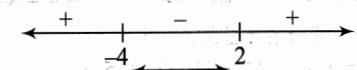
$$\Rightarrow x \leq -2 \text{ and } x \geq 1$$

$$\Rightarrow x \in (-\infty, -2) \cup (1, \infty)$$

134. Given in-equation is

$$x^2 + 2x - 8 < 0$$

$$\Rightarrow (x+4)(x-2) < 0$$



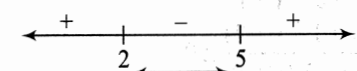
$$\Rightarrow -4 < x < 2$$

$$\Rightarrow x \in (-4, 2)$$

135. Given in-equation is

$$x^2 - 7x + 10 \leq 0$$

$$\Rightarrow (x-2)(x-5) \leq 0$$



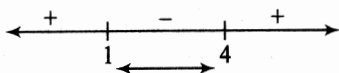
$$\Rightarrow 2 \leq x \leq 5$$

$$\Rightarrow x \in [2, 5]$$

143. Given in-equation is

$$-x^2 + 5x - 4 > 0$$

$$\Rightarrow x^2 - 5x + 4 < 0$$



$$\Rightarrow (x-1)(x-4) < 0$$

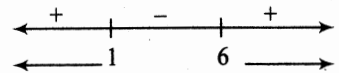
$$\Rightarrow 1 < x < 4$$

144. Given in-equation is

$$-x^2 + 7x - 6 \leq 0$$

$$\Rightarrow x^2 - 7x + 6 \geq 0$$

$$\Rightarrow (x-1)(x-6) \geq 0$$



$$\Rightarrow x \leq 1, x \geq 6$$

$$\Rightarrow x \in (-\infty, 1] \cup [6, \infty)$$

152. Given in-equation is

$$\frac{1}{x} > 2$$

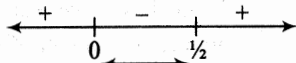
$$\Rightarrow \frac{1}{x} - 2 > 0$$

$$\Rightarrow \frac{1-2x}{x} > 0$$

$$\Rightarrow \frac{2x-1}{x} > 0$$

$$\Rightarrow 0 < x < \frac{1}{2}$$

$$\Rightarrow x \in \left(0, \frac{1}{2}\right)$$



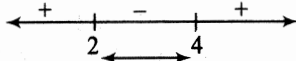
153. Given in-equation is

$$\frac{x-2}{4-x} \geq 0$$

$$\Rightarrow \frac{x-2}{x-4} \leq 0$$

$$\Rightarrow 2 \leq x \leq 4 \text{ and } x \neq 4$$

$$\Rightarrow x \in [2, 4)$$

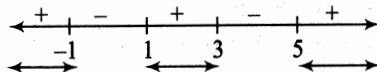


154. Given in-equation is

$$\frac{(x-1)(x-3)}{(x+1)(x-5)} \geq 0$$

$$\Rightarrow x < -1, 1 \leq x \leq 3, x > 5$$

$$\Rightarrow x \in (-\infty, -1) \cup [1, 3] \cup (5, \infty)$$



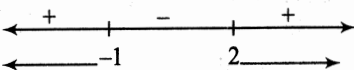
155. Given in-equation is

$$\frac{1}{x-2} \geq \frac{1}{x+1}$$

$$\Rightarrow \frac{1}{x-2} - \frac{1}{x+1} \geq 0$$

$$\Rightarrow \frac{(x+1)-(x-2)}{(x-2)(x+1)} \geq 0$$

$$\Rightarrow \frac{3}{(x-2)(x+1)} \geq 0$$



$$\Rightarrow x < -1, x > 2$$

$$\Rightarrow x \in (-\infty, -1) \cup (2, \infty)$$

164. Given in-equation is

$$x^2(x-2)^4(x-3) \geq 0$$

$$\Rightarrow (x-3) \geq 0$$

$$\Rightarrow x \geq 3$$

$$\Rightarrow x \in [3, \infty)$$

165. Given in-equation is

$$(x-1)^4(x-3)^6(x-2)^5(x-4)^{10} > 0$$

$$\Rightarrow (x-2)^5 > 0$$

$$\Rightarrow (x-2) > 0$$

$$\Rightarrow x > 2, \text{ provided } x \neq 1, 3, 4$$

Hence the solution set is $x \in (2, \infty) - \{3, 4\}$.

166. Given in-equation is

$$\frac{(x-1)^{2013}(x-2)^{2014}}{(x-3)^{2016}(x-5)^{2010}} > 0$$

$$\Rightarrow (x-1)^{2013} > 0$$

$$\Rightarrow (x-1) > 0$$

$$\Rightarrow x > 1, \text{ provided } x \neq 2, 3, 5$$

$$\Rightarrow x \in (1, \infty) \cup \{2, 3, 5\}$$

174. Given in-equation is $x^2 + x + 2 > 0$.

This is true for every values of x .
Hence, $x \in \mathbb{R}$

175. Given in-equation is $x^2 - x + 3 < 0$.

This statement is false for every values of x .
Hence, $x = \emptyset$

176. Given in-equation is

$$(x+2)(x^4 + x^2 + 1) > 0$$

$$\Rightarrow (x+2)(x^2 - x + 1)(x^2 - x + 1) > 0$$

$$\Rightarrow (x+2) > 0$$

$$\Rightarrow x > -2$$

$$\Rightarrow x \in (-2, \infty)$$

177. Given in-equation is

$$x^4 - 4 \leq 0$$

$$\Rightarrow (x^2 - 2)(x^2 + 2) \leq 0$$

$$\Rightarrow (x^2 - 2) \leq 0$$

$$\Rightarrow (x + \sqrt{2})(x - \sqrt{2}) \leq 0$$

$$\Rightarrow -\sqrt{2} \leq x \leq \sqrt{2}$$

$$\Rightarrow x \in [-\sqrt{2}, \sqrt{2}]$$

178. Given in-equation is

$$(x^2 + 4x + 1)(x^2 + 1) \geq 0$$

$$\Rightarrow (x^2 + 4x + 1) \geq 0$$

$$\Rightarrow (x + 2 + \sqrt{3})(x + 2 - \sqrt{3}) \geq 0$$

$$\Rightarrow (x + 2 + \sqrt{3}) \leq 0, (x + 2 - \sqrt{3}) \geq 0$$

$$\Rightarrow x \leq -(2 + \sqrt{3}), x \geq -(2 - \sqrt{3})$$

$$\Rightarrow x \in (-\infty, -(2 + \sqrt{3})] \cup [-(2 - \sqrt{3}), \infty)$$

188. Now, $x^2 - 3x + 2 > 0$

$$\Rightarrow (x-1)(x-2) > 0$$

$$\Rightarrow x < 1 \text{ and } x > 2$$

$$\Rightarrow x \in (-\infty, 1) \cup (2, \infty) \quad \dots(i)$$

Also, $x^2 + 2x - 8 < 0$

$$\Rightarrow (x+4)(x-2) < 0$$

$$\Rightarrow -4 < x < 2$$

$$\Rightarrow x \in (4, 2) \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$x \in (-4, 1)$$

189. Now, $x^2 - 9 \leq 0$
 $\Rightarrow (x+3)(x-3) \leq 0$
 $\Rightarrow -3 \leq x \leq 3$... (i)

Also, $x^2 - 1 \geq 0$
 $\Rightarrow (x+1)(x-1) \geq 0$
 $\Rightarrow x \leq -1, x \geq 1$
 $\Rightarrow x \in (-\infty, -1] \cup [1, \infty)$... (ii)

From Eqs (i) and (ii), we get
 $x \in [-3, -1] \cup [1, 3]$

190. Now, $x^3 - 9x \geq 0$
 $\Rightarrow x(x^2 - 9) \geq 0$
 $\Rightarrow x(x+3)(x-3) \geq 0$
 $\Rightarrow -3 \leq x \leq 0, x \geq 3$... (i)

Also, $x^3 + 4x \leq 0$
 $\Rightarrow x(x^2 + 4) \leq 0$
 $\Rightarrow x \leq 0$... (ii)

From Eqs (i) and (ii), we get
 $x \in [-3, 0]$

198. We have,
 $|x|^3 - 3|x| + 2 = 0$
 $\Rightarrow (|x| - 1)(|x| - 2) = 0$
 $\Rightarrow (|x| - 1) = 0, (|x| - 2) = 0$
 $\Rightarrow |x| = 1, |x| = 2$
 $\Rightarrow x = \pm 1, x = \pm 2$

199. We have,
 $x^2 - 4|x| + 3 = 0$
 $\Rightarrow |x|^3 - 4|x| + 3 = 0$
 $\Rightarrow (|x| - 1)(|x| - 3) = 0$
 $\Rightarrow (|x| - 1) = 0, (|x| - 3) = 0$
 $\Rightarrow |x| = 1, |x| = 3$
 $\Rightarrow x = \pm 1, x = \pm 3$
 Hence the sum of the roots $= 1 - 1 + 3 - 3 = 0$

200. We have,
 $|3x - 1| = |x + 5|$
 $\Rightarrow (3x - 1) = \pm(x + 5)$
 Taking positive sign, we get
 $3x - 1 = x + 5$
 $\Rightarrow 2x = 6$
 $\Rightarrow x = 3$
 Taking negative sign, we get
 $3x - 1 = -x - 5$
 $\Rightarrow 4x = -4$
 $\Rightarrow x = -1$

Hence, the solutions are $x = -1, 3$.

201. We have,
 $|2x - 5| = x - 3$
 $\Rightarrow (x^2 - x - 6) = \pm(x + 2)$
 Taking positive sign, we get
 $2x - 5 = x - 3$
 $\Rightarrow 2x - x = 5 - 3$
 $\Rightarrow x = 2$
 Taking negative sign, we get
 $2x - 5 = -x + 3$
 $\Rightarrow 2x + x = 5 + 3$
 $\Rightarrow 3x = 8$
 $\Rightarrow x = 8/3$
 Hence, the solution set is $\{2, 8/3\}$.

202. We have,
 $|x^2 - x - 6| = x + 2$
 $\Rightarrow (x^2 - x - 6) = \pm(x + 2)$
 Taking positive sign, we get
 $(x^2 - x - 6) = (x + 2)$
 $\Rightarrow (x^2 - 2x - 8) = 0$
 $\Rightarrow (x - 4)(x + 2) = 0$
 $\Rightarrow x = -2, 4$.

Taking negative sign, we get
 $(x^2 - x - 6) = -(x + 2)$
 $\Rightarrow (x^2 - 4) = 0$
 $\Rightarrow x = \pm 2$

Hence the solution set is $\{-2, 2, 4\}$.

203. We have $2|x - 2| + 3|x - 4| = 3$
Case I: When $x < 2$
 $\Rightarrow -2(x - 2) - 3(x - 4) = 3$
 $\Rightarrow -2x + 4 - 3x + 12 = 3$
 $\Rightarrow -5x = 3 - 16 = -13$
 $\Rightarrow x = 13/5$

It is rejected, since $x = 13/5 > 2$.

Case II: When $2 \leq x \leq 4$
 $\Rightarrow 2(x - 2) - 3(x - 4) = 3$
 $\Rightarrow 2x - 4 - 3x + 12 = 3$
 $\Rightarrow -x = 3 - 8$
 $\Rightarrow x = 5$

It is rejected, since $x = 5$ not lies in the given interval.

Case III: When $x \geq 4$
 $\Rightarrow 2(x - 2) + 3(x - 4) = 3$
 $\Rightarrow 2x - 4 + 3x - 12 = 3$
 $\Rightarrow 5x = 3 + 4 + 12 = 19$
 $\Rightarrow x = 19/5$

It is rejected, since $x = 19/5 < 4$.

Hence, the solution set is ϕ .

204. We have $|x| + |x - 2| = 4$
Case I: When $x < 0$.
 $\Rightarrow -x - (x - 2) = 4$
 $\Rightarrow -x - x + 2 = 4$
 $\Rightarrow -2x = 4 - 2 = 2$
 $\Rightarrow x = -1$

Case II: When $0 \leq x \leq 2$
 $\Rightarrow x - (x - 2) = 4$
 $\Rightarrow 2 = 4$

It is not possible.

Case III: When $x \geq 2$
 $\Rightarrow x + x - 2 = 4$
 $\Rightarrow 2x = 4 + 2$
 $\Rightarrow 2x = 6$
 $\Rightarrow x = 3$

Hence, the solution set is $\{-1, 3\}$.

205. We have,
 $|x - 1| + |x - 3| = 2$
 $\Rightarrow |x - 1| + |3 - x| = 2$
 $\Rightarrow (x - 1)(3 - x) \geq 0$
 $\Rightarrow (x - 1)(x - 3) \leq 0$
 $\Rightarrow 1 \leq x \leq 3$
 $\Rightarrow x \in [1, 3]$

206. We have,
 $|x^2 - 1| + |x^2 - 4| = 3$

$$\begin{aligned} \Rightarrow |x^2 - 1| + |4 - x^2| &= 3 \\ \Rightarrow (x^2 - 1)(4 - x^2) &\geq 0 \\ \Rightarrow (x^2 - 1)(x^2 - 4) &\leq 0 \\ \Rightarrow 1 \leq x^2 \leq 4 \\ \Rightarrow 1 \leq \sqrt{x^2} \leq 2 \\ \Rightarrow 1 \leq |x| \leq 2 \\ \Rightarrow x \in [-2, -1] \cup [1, 2] \end{aligned}$$

207. We have,

$$\begin{aligned} \sqrt{x+2\sqrt{x-1}} + \sqrt{x-2\sqrt{x-1}} &= 2 \\ \text{Put } x-1 &= t^2 \\ \Rightarrow \sqrt{t^2+1+2t} + \sqrt{t^2+1-2t} &= 2 \\ \Rightarrow \sqrt{(t+1)^2} + \sqrt{(t-1)^2} &= 2 \\ \Rightarrow |t+1| + |t-1| &= 2 \\ \Rightarrow |t+1| + |1-t| &= 2 \\ \Rightarrow (t+1)(1-t) &\geq 0 \\ \Rightarrow (t+1)(t-1) &\leq 0 \\ \Rightarrow -1 \leq t \leq 1 \\ \Rightarrow 0 \leq t^2 \leq 1 \\ \Rightarrow 0 \leq x-1 \leq 1 \\ \Rightarrow 1 \leq x \leq 2 \\ \Rightarrow x \in [1, 2] \end{aligned}$$

208. We have,

$$\begin{aligned} \left| \frac{x}{x-1} \right| + |x| &= \frac{x^2}{|x-1|} \\ \Rightarrow \left| \frac{x}{x-1} \right| + |x| &= \left| \frac{x}{x-1} + x \right| \\ \Rightarrow x \cdot \left(\frac{x}{x-1} \right) &\geq 0 \\ \Rightarrow \left(\frac{x^2}{x-1} \right) &\geq 0 \\ \Rightarrow \left(\frac{1}{x-1} \right) &\geq 0 \\ \Rightarrow x &> 1 \\ \Rightarrow x \in (1, \infty) \end{aligned}$$

209. We have,

$$\begin{aligned} 2^{|x+1|} - 2^x &= |2^x - 1| + 1 \\ \Rightarrow |2^x - 1| + (2^x + 1) &= 2^{|x+1|} \\ \Rightarrow |2^x - 1| + |2^x + 1| &= 2^{|x+1|} \\ \Rightarrow (2^x - 1)(2^x + 1) &\geq 0 \\ \Rightarrow (2^{2x} - 1) &\geq 0 \\ \Rightarrow 2^{2x} &\geq 1 \\ \Rightarrow 2^{2x} &\geq 2^0 \\ \Rightarrow x &\geq 0 \\ \Rightarrow x \in [0, \infty) \end{aligned}$$

210. We have,

$$\begin{aligned} |x^2 + x - 20| &= -(x^2 + x - 20) \\ \Rightarrow (x^2 + x - 20) &\leq 0 \\ \Rightarrow (x+5)(x-4) &\leq 0 \\ \Rightarrow -5 \leq x \leq 4 \end{aligned}$$

211. We have,

$$\left| \frac{x^2 - 6x + 8}{x^2 - 4x + 3} \right| = - \left(\frac{x^2 - 6x + 8}{x^2 - 4x + 3} \right)$$

$$\begin{aligned} \Rightarrow \left(\frac{x^2 - 6x + 8}{x^2 - 4x + 3} \right) &\leq 0 \\ \Rightarrow \left(\frac{(x-2)(x-4)}{(x-1)(x-3)} \right) &\leq 0 \\ \Rightarrow 1 < x \leq 2, 3 < x \leq 4 \\ \Rightarrow x \in (1, 2] \cup (3, 4] \end{aligned}$$

212. We have,

$$\begin{aligned} |x-1| &< 3 \\ \Rightarrow -3 < (x-1) &< 3 \\ \Rightarrow -3+1 < x < 3+1 \\ \Rightarrow -2 < x < 4 \\ \Rightarrow x \in (-2, 4) \end{aligned}$$

213. We have,

$$\begin{aligned} |x-4| &\leq 3 \\ \Rightarrow -3 \leq (x-4) &\leq 3 \\ \Rightarrow -3+4 \leq x \leq 3+4 \\ \Rightarrow 1 \leq x \leq 7 \\ \Rightarrow x \in [1, 7] \end{aligned}$$

214. We have,

$$\begin{aligned} |x| > 5 \\ \Rightarrow (x+2) > 5, (x+2) < -5 \\ \Rightarrow x > 3, x < -7 \\ \Rightarrow x \in (-\infty, -7] \cup (3, \infty) \end{aligned}$$

215. We have,

$$\begin{aligned} |x-2| &\geq 3 \\ \Rightarrow (x-2) \geq 3, (x-2) &\leq -3 \\ \Rightarrow x \geq 5, x \leq -1 \\ \Rightarrow x \in (-\infty, -1] \cup [5, \infty) \end{aligned}$$

216. We have,

$$\begin{aligned} \left| \frac{4x-2}{3} \right| &\leq 2 \\ \Rightarrow -2 \leq \frac{4x-2}{3} &\leq 2 \\ \Rightarrow -6 \leq 4x-2 &\leq 6 \\ \Rightarrow -6+2 \leq 4x &\leq 6+2 \\ \Rightarrow -4 \leq 4x &\leq 8 \\ \Rightarrow -1 \leq x &\leq 2 \\ \Rightarrow x \in [-1, 2] \end{aligned}$$

217. We have,

$$x^2 - |3x-2| > 0$$

Case I: When $x \geq 3/2$

Then $x^2 - 3x + 2 > 0$

$$\begin{aligned} \Rightarrow (x-1)(x-2) &> 0 \\ \Rightarrow x < 1, x > 2 \\ \Rightarrow x &> 2 \end{aligned}$$

Case II: When $x < 3/2$

Then $x^2 + 3x + 2 > 0$

$$\begin{aligned} \Rightarrow (x+1)(x+2) &> 0 \\ \Rightarrow x < -2 \text{ and } x > -1 \end{aligned}$$

Hence the solution set is

$$(-\infty, -2) \cup (1, 2) \cup (2, \infty)$$

218. We have,

$$|x| + |x-2| > 3$$

Case I: When $x < 0$

Then $-x - x + 2 > 3$

$$\begin{aligned} \Rightarrow -2x &> 1 \\ \Rightarrow x &< -1/2 \end{aligned}$$

Case II: When $x \geq 2$

Then $x + x - 2 > 3$

$$\Rightarrow 2x > 5$$

$$\Rightarrow x > 5/2$$

Hence the solution set is $(-\infty, -1/2) \cup (5/2, \infty)$.

219. We have,

$$|x + 2| + |x| + |x - 2| > 6$$

Case I: When $x < -2$

Then $-x - 2 - x - x + 2 > 6$

$$\Rightarrow -3x > 6$$

$$\Rightarrow x < -2$$

Case II: When $x \geq 2$

Then $x + 2 + x + x - 2 > 6$

$$\Rightarrow 3x > 6$$

$$\Rightarrow x > 2$$

Hence the solution set is $(-\infty, -2) \cup (2, \infty)$.

220. We have,

$$\left| \frac{2}{x+3} \right| \leq 1$$

$$\Rightarrow \frac{2}{|x+3|} \leq 1$$

$$\Rightarrow |x+3| \geq 2$$

$$\Rightarrow x+3 \geq 2, x+3 \leq -2$$

$$\Rightarrow x \geq -1, x \leq -5$$

$$\Rightarrow x \in (-\infty, -5] \cup [-1, \infty)$$

Hence the solution set is $(-\infty, -5] \cup [-1, \infty)$.

221. We have,

$$|3x+2| > |2x-1|$$

$$\Rightarrow |3x+2|^2 > |2x-1|^2$$

$$\Rightarrow (3x+2)^2 > (2x-1)^2$$

$$\Rightarrow (3x+2)^2 - (2x-1)^2 > 0$$

$$\Rightarrow ((3x+2) + (2x-1))((3x+2) - (2x-1)) > 0$$

$$\Rightarrow (5x+1)(x+3) > 0$$

$$\Rightarrow x < -3, x > -1/5$$

$$\Rightarrow x \in (-\infty, -3) \cup (-1/5, \infty)$$

Hence the solution set is $(-\infty, -3) \cup (-1/5, \infty)$.

222. We have,

$$\sqrt{(2x+7)} + \sqrt{(x+4)} = 0$$

The LHS of the equation is defined for all positive real numbers whereas RHS is zero.

So, the equation has no real roots.

223. We have,

$$\sqrt{(x-4)} = -5$$

The LHS of the equation is defined for all positive real numbers whereas RHS is negative.

So, the equation has no real roots.

224. We have,

$$\sqrt{(x-6)} - \sqrt{(8-x)} = 2$$

The LHS of the equation is defined when $x \geq 6, x \leq 8$ i.e. $6 \leq x \leq 8$ whereas RHS is 2.

So, the equation has no real roots.

225. We have,

$$\sqrt{(-2-x)} = \sqrt[3]{(x-7)}$$

The LHS of the equation is defined for $x \leq -2$ and its value is zero at $x = -2$ and positive $x < -2$ whereas

RHS of the equation is defined for all real values and its value is zero at $x = 7$, positive at $x > 7$ and negative at $x < 7$.

So, the equation has no real roots.

226. The L.H.S of the given equation is ≥ 4 for $x \geq 0$ whereas the RHS is 3

So, the equation has no real roots.

227. We have,

$$7\sqrt{x} + 8\sqrt{-x} + \frac{15}{x^3} = 98.$$

The expression $7\sqrt{x} + 8\sqrt{-x}$ of the given equation is defined only when at $x = 0$ and the expression $\frac{15}{x^3}$ is undefined at $x = 0$.

So the LHS of the given equation is not satisfied any real value of x , whereas RHS. is 98.

Therefore, the given equation has no real roots.

228. We have,

$$\sqrt{x-2} + \sqrt{4-x} = \sqrt{6-x}$$

$$\Rightarrow x-2+4-x+2\sqrt{x-2}\sqrt{4-x} = 6-x$$

$$\Rightarrow 2\sqrt{x-2}\sqrt{4-x} = 4-x$$

$$\Rightarrow 4(x-2)(4-x) = (4-x)^2$$

$$\Rightarrow (4-x)(4x-8-4+x) = 0$$

$$\Rightarrow (4-x)(5x-12) = 0$$

$$\Rightarrow x = 4, 12/5$$

Hence the solution is $\{4, 12/5\}$.

229. We have,

$$\sqrt{2x-4} - \sqrt{x+5} = 1$$

$$\Rightarrow \sqrt{2x-4} = 1 + \sqrt{x+5}$$

$$\Rightarrow 2x-4 = 1+x+5+2\sqrt{x+5}$$

$$\Rightarrow x-10 = 2\sqrt{x+5}$$

$$\Rightarrow (x-10)^2 = 4(x+5)$$

$$\Rightarrow x^2 - 20x + 100 = 4x + 20$$

$$\Rightarrow x^2 - 24x + 80 = 0$$

$$\Rightarrow (x-4)(x-20) = 0$$

$$\Rightarrow x = 4, 20$$

Here, $x = 4$ is not satisfied the given equation.

Hence the solution set is $x = 20$.

230. We have,

$$\sqrt{3x+4} + \sqrt{x-4} = 2\sqrt{x}$$

$$\Rightarrow 3x+4+x-4+2\sqrt{3x+4}\sqrt{x-4} = 4x$$

$$\Rightarrow 2\sqrt{3x+4}\sqrt{x-4} = 0$$

$$\Rightarrow x = -4/3, 4$$

Here, $x = 4/3$ does not satisfy the given equation. Hence the solution set is $x = 4$.

231. We have,

$$\sqrt{x-1} + \sqrt{2x+6} = 6$$

$$\Rightarrow x-1+2x+6+2\sqrt{x-1}\sqrt{2x+6} = 36$$

$$\Rightarrow 2\sqrt{(x-1)(2x+6)} = -3x+31$$

$$\Rightarrow 4(2x^2+4x-6) = (3x-31)^2$$

$$\Rightarrow 4(2x^2+4x-6) = 9x^2-186x+961$$

$$\Rightarrow x^2-202x+985=0$$

$$\Rightarrow (x-5)(x-197)=0$$

$$\Rightarrow x=5, 197$$

Here, $x=197$ does not satisfy the given equation.

Hence, the solution set is $x=5$.

232. We have,

$$\sqrt[5]{\frac{3}{x+1} + \frac{7}{x+2}} < \sqrt[5]{\frac{6}{x-1}}$$

$$\Rightarrow \frac{3}{x+1} + \frac{7}{x+2} < \frac{6}{x-1}$$

$$\Rightarrow \frac{3x+6+7x+7}{(x+1)(x+2)} - \frac{6}{x-1} < 0$$

$$\Rightarrow \frac{10x+13}{(x+1)(x+2)} - \frac{6(x^2+3x+2)}{x-1} < 0$$

$$\Rightarrow \frac{10x^2+3x-13-6x^2-18x-12}{(x+1)(x+2)(x-1)} < 0$$

$$\Rightarrow \frac{4x^2-15x-25}{(x+1)(x+2)(x-1)} < 0$$

$$\Rightarrow \frac{(x-5)(4x+5)}{(x+1)(x+2)(x-1)} < 0$$

$$\Rightarrow x < -2, -5/4 < x < -1, 1 < x < 5$$

Hence the solution set is

$$(-\infty, -2) \cup (-5/4, -1) \cup (1, 5).$$

233. We have,

$$\sqrt[3]{\frac{x-2}{x-1}} < \sqrt[3]{\frac{1}{x-1}}$$

$$\Rightarrow \frac{x-2}{x-1} - \frac{1}{x-1} < 0$$

$$\Rightarrow \frac{x-2-1}{x-1} < 0$$

$$\Rightarrow \frac{x-3}{x-1} < 0$$

$$\Rightarrow 1 < x < 3$$

Hence the solution set is $(1, 3)$.

234. We have,

$$\sqrt{3x-2} < \sqrt{x+4}$$

$$\Rightarrow \begin{cases} 3x-2 \geq 0 \\ x+4 > 0 \\ x+4 > 3x-2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 2/3 \\ x > -4 \\ x < 6 \end{cases}$$

Hence, the solution set is $\left[\frac{2}{3}, 6\right)$.

235. We have,

$$\sqrt{4x-3} < \sqrt{2x+5}$$

$$\Rightarrow \begin{cases} 4x-3 \geq 0 \\ 2x+5 > 0 \\ 2x+5 > 4x+3 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 3/4 \\ x > -5/2 \\ x < 1 \end{cases}$$

Hence the solution set is $\left[\frac{3}{4}, 1\right)$.

236. We have $\sqrt[3]{(3x-5)} < (x-1)$

$$\Rightarrow (3x-5) < (x-1)^3$$

$$\Rightarrow x^3 - 3x^2 + 3x - 1 > 3x - 5$$

$$\Rightarrow x^3 - 3x^2 + 4 > 0$$

$$\Rightarrow x^3 + x^2 - 4x^2 - 4x + 4x + 4 > 0$$

$$\Rightarrow x^2(x+1) - 4x(x+1) + 4(x+1) > 0$$

$$\Rightarrow (x+1)(x^2 - 4x + 4) > 0$$

$$\Rightarrow (x+1)(x-2)^2 > 0$$

$$\Rightarrow x+1 > 0$$

$$\Rightarrow x > -1$$

Hence the solution set is $(-1, \infty) - \{2\}$

237. We have,

$$\sqrt[3]{3x-2} < x$$

$$\Rightarrow 3x-2 < x^3$$

$$\Rightarrow x^3 - 3x + 2 > 0$$

$$\Rightarrow x^3 - x^2 + x^2 - x - 2x + 2 > 0$$

$$\Rightarrow x^2(x-1) + x(x-1) - 2(x-1) > 0$$

$$\Rightarrow (x-1)(x^2 + x - 2) > 0$$

$$\Rightarrow (x-1)(x-1)(x+2) > 0$$

$$\Rightarrow (x-1)^2(x+2) > 0$$

$$\Rightarrow x+2 > 0$$

$$\Rightarrow x > -2$$

Hence the solution set is $(-2, 1) \cup (1, \infty)$.

238. We have,

$$\sqrt{(x+14)} < (x+2)$$

$$\Rightarrow \begin{cases} x+14 \geq 0 \\ x+2 > 0 \end{cases}$$

$$\Rightarrow \begin{cases} x+2 > 0 \\ (x+2)^2 > x+14 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

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$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq -14 \\ x > -2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 1 \\ x > 1 \\ x^2 - 4x + 3 > 0 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 1 \\ x > 1 \\ (x-1)(x-3) > 0 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 1 \\ x > 1 \\ x < 1, x > 3 \end{cases}$$

Hence the solution set is $(3, \infty)$.

240. We have,

$$\Rightarrow \sqrt[3]{5x-4} > x$$

$$\Rightarrow 5x-4 > x^3$$

$$\Rightarrow 5x-4-x^3 > 0$$

$$\Rightarrow x^3-5x+4 > 0$$

$$\Rightarrow x^3-x^2+x^2-x-4x+4 > 0$$

$$\Rightarrow x^2(x-1)+x(x-1)-4(x-1) > 0$$

$$\Rightarrow (x-1)(x^2+x-4) > 0$$

$$\Rightarrow \frac{-1-\sqrt{17}}{2} < x < 1, x > \frac{-1+\sqrt{17}}{2}$$

Hence the solution set is

$$\left(\frac{-1-\sqrt{17}}{2}, 1 \right) \cup \left(\frac{-1+\sqrt{17}}{2}, \infty \right)$$

241. We have,

$$\Rightarrow \sqrt[3]{x^3-7x^2} > (-36)^{1/3}$$

$$\Rightarrow x^3-7x^2 > -36$$

$$\Rightarrow x^3-7x^2+36 > 0$$

$$\Rightarrow x^3+2x^2-9x^2-18x+18x+36 > 0$$

$$\Rightarrow x^2(x+2)-9x(x+2)+18(x+2) > 0$$

$$\Rightarrow (x+2)(x^2-9x+18) > 0$$

$$\Rightarrow (x+2)(x^2-9x+18) > 0$$

$$\Rightarrow (x+2)(x-3)(x-6) > 0$$

$$\Rightarrow -2 < x < 3, x > 6$$

Hence the solution set is $(-2, 3) \cup (6, \infty)$.

242. We have,

$$\Rightarrow \sqrt{-x^2+4x-3} > (6-2x)$$

$$\Rightarrow \begin{cases} -x^2+4x-3 > (6-2x)^2 \\ 6-2x \geq 0 \end{cases} \text{ and } \begin{cases} -x^2+4x-3 \geq 0 \\ 6-2x < 0 \end{cases}$$

$$\Rightarrow \begin{cases} 5x^2-28x+39 < 0 \\ 6-2x \geq 0 \end{cases} \text{ and } \begin{cases} x^2-4x+3 \leq 0 \\ 3-x < 0 \end{cases}$$

$$\Rightarrow \begin{cases} 5x^2-15x-13x+39 < 0 \\ 6-2x \geq 0 \end{cases} \text{ and } \begin{cases} (x-1)(x-3) \leq 0 \\ x-3 > 0 \end{cases}$$

$$\Rightarrow \begin{cases} (x-3)(5x-13) < 0 \\ 6-2x \geq 0 \end{cases} \text{ and } \begin{cases} 1 \leq x \leq 3 \\ x > 3 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{13}{5} < x < 3 \\ x \leq 3 \end{cases} \text{ and } \begin{cases} 1 \leq x \leq 3 \\ x > 3 \end{cases}$$

Hence the solution set is $\left(\frac{13}{5}, 3 \right)$.

243. We have,

$$\Rightarrow \sqrt{x-2} + \sqrt{x-1} > 2$$

$$\Rightarrow \begin{cases} x-2 \geq 0 \\ x-1 \geq 0 \\ x-2+x-1+2\sqrt{x-2}\sqrt{x-1} > 4 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 2 \\ x \geq 1 \\ 2\sqrt{x-2}\sqrt{x-1} > (7-2x) \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 2 \\ x \geq 1 \\ 4(x^2-3x+2) > (7-2x)^2 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 2 \\ x \geq 1 \\ 16x > 41 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 2 \\ x \geq 1 \\ x > 41/16 \end{cases}$$

Hence, the solution set is $\left(\frac{41}{16}, \infty \right)$.

244. We have,

$$\Rightarrow \sqrt{x-6} - \sqrt{10-x} \geq 1$$

$$\Rightarrow \begin{cases} x-6 \geq 0 \\ 10-x \leq 0 \\ \sqrt{x-6} \geq 1 + \sqrt{10-x} \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 6 \\ 10 \leq x \\ x-6 \geq 1+10-x+2\sqrt{10-x} \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 6 \\ 10 \leq x \\ (2x-17)^2 \geq 4(10-x) \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 6 \\ 10 \leq x \\ 4x^2-68x+289 \geq 40-4x \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 6 \\ 10 \leq x \\ 4x^2-64x+249 \geq 0 \end{cases}$$

$$\Rightarrow \begin{cases} x \geq 6 \\ 10 \leq x \\ x \leq \left(8 - \frac{\sqrt{7}}{2}\right), x \geq \left(8 + \frac{\sqrt{7}}{2}\right) \end{cases}$$

Hence the solution set is $\left[8 + \frac{\sqrt{7}}{2}, 10\right]$

245. We have,

$$\begin{aligned} 5^{x^2+3x+2} &= 5^0 \\ \Rightarrow x^2 + 3x + 2 &= 0 \\ \Rightarrow (x+1)(x+2) &= 0 \\ \Rightarrow x &= -1, -2 \end{aligned}$$

Hence the solution set is $\{-1, -2\}$.

246. We have,

$$\begin{aligned} 3^{x^2+5|x|+6} &= 1 \\ \Rightarrow 3^{x^2+5|x|+6} &= 3^0 \\ \Rightarrow x^2 + 5|x| + 6 &= 0 \\ \Rightarrow |x|^2 + 5|x| + 6 &= 0 \\ \Rightarrow (|x|+2)(|x|+3) &= 0 \\ \Rightarrow |x| &= -2, -3 \\ \Rightarrow x &= \emptyset \end{aligned}$$

Hence the solution set is $\emptyset$.

247. We have,

$$\begin{aligned} 5^{2x} - 24 \cdot 5^x - 25 &= 0 \\ \Rightarrow (5^x)^2 - 24(5^x) - 25 &= 0 \\ \Rightarrow a^2 - 24a - 25 &= 0, a = 5^x \\ \Rightarrow (a-25)(a+1) &= 0 \\ \Rightarrow a &= 25, -1 \end{aligned}$$

When $a = 25 \Rightarrow 5x = 5^2 \Rightarrow x = 2$

When $a = -1 \Rightarrow 5x = -1 \Rightarrow x = \emptyset$

Hence the solution set is $\{2\}$.

248. We have,

$$\begin{aligned} 64 \cdot 9^x - 84 \cdot 12^x + 27 \cdot 16^x &= 0 \\ \Rightarrow 64 \left(\frac{9}{16}\right)^x - 84 \left(\frac{12}{16}\right)^x + 27 &= 0 \\ \Rightarrow 64 \left(\frac{3}{4}\right)^{2x} - 84 \left(\frac{3}{4}\right)^x + 27 &= 0 \\ \Rightarrow 64a^2 - 84a + 27 &= 0, a = \left(\frac{3}{4}\right)^x \\ \Rightarrow 64a^2 - 48a - 36a + 27 &= 0 \\ \Rightarrow 16a(4a-3) - 9(4a-3) &= 0 \\ \Rightarrow (4a-3)(16a-9) &= 0 \\ \Rightarrow a &= 3/4, 9/16 \end{aligned}$$

When $a = \frac{3}{4} \Rightarrow \left(\frac{3}{4}\right)^x = \left(\frac{3}{4}\right) \Rightarrow x = 1$

When $a = \frac{9}{16} \Rightarrow \left(\frac{3}{4}\right)^x = \left(\frac{3}{4}\right)^2 \Rightarrow x = 2$

Hence the solution set is $\{1, 2\}$.

249. We have,

$$\begin{aligned} 15 \cdot 2^{x+1} + 15 \cdot 2^{2-x} &= 135 \\ \Rightarrow 30 \cdot 2^x + \frac{60}{2^x} &= 135 \end{aligned}$$

$$\begin{aligned} \Rightarrow 30 \cdot a + \frac{60}{a} &= 135, a = 2^x \\ \Rightarrow 30a^2 - 135a + 60 &= 0 \\ \Rightarrow 6a^2 - 27a + 12 &= 0 \\ \Rightarrow 2a^2 - 9a + 4 &= 0 \\ \Rightarrow 2a^2 - 8a + a + 4 &= 0 \\ \Rightarrow 2a(a-4) - 1(a-4) &= 0 \\ \Rightarrow (a-4)(2a-1) &= 0 \\ \Rightarrow a &= 4, 1/2 \end{aligned}$$

When $a = 4 \Rightarrow 2^x = 4 \Rightarrow 2^x = 2^2 \Rightarrow x = 2$

When $a = \frac{1}{2} \Rightarrow 2^x = 2^{-1} \Rightarrow x = -1$

250. We have,

$$\begin{aligned} 3^{x-4} + 5^{x-4} &= 34 \\ \Rightarrow 3^{x-4} + 5^{x-4} &= 3^2 + 5^2 \\ \Rightarrow x-4 &= 2 \\ \Rightarrow x &= 6 \end{aligned}$$

Hence the solution is $x = 6$

251. We have,

$$\begin{aligned} 1 + 3^{x/2} &= 2^x \\ \Rightarrow 1^x + (\sqrt{3})^x &= 2^x \\ \Rightarrow x &= 2 \end{aligned}$$

Hence the solution is $x = 2$.

252. We have,

$$\begin{aligned} 3^x + 4^x + 5^x &= 6^x \\ \Rightarrow \left(\frac{3}{6}\right)^x + \left(\frac{4}{6}\right)^x + \left(\frac{5}{6}\right)^x &= 1 \\ \Rightarrow x &= 3 \end{aligned}$$

Hence the solution is $x = 3$.

253. We have,

$$\begin{aligned} 1^x + 6^x + 8^x &= 9^x \\ \Rightarrow \left(\frac{1}{9}\right)^x + \left(\frac{6}{9}\right)^x + \left(\frac{8}{9}\right)^x &= 1 \\ \Rightarrow x &= 3 \end{aligned}$$

Hence the solution set is $x = 3$

254. We have,

$$\begin{aligned} 16^{\sin^2 x} + 16^{\cos^2 x} &= 10 \\ \Rightarrow 16^{\sin^2 x} + 16^{1-\sin^2 x} &= 10 \\ \Rightarrow 16^{\sin^2 x} + \frac{16}{16^{\sin^2 x}} &= 10 \\ \Rightarrow a + \frac{16}{a} &= 10, \text{ where } a = 16^{\sin^2 x} \\ \Rightarrow a^2 - 10a + 16 &= 0 \\ \Rightarrow (a-2)(a-8) &= 0 \\ \Rightarrow a &= 2, 8 \end{aligned}$$

When $a = 2 \Rightarrow 16^{\sin^2 x} = 2$

$$\Rightarrow 2^{4\sin^2 x} = 2$$

$$\Rightarrow 4\sin^2 x = 1$$

$$\Rightarrow \sin^2 x = \left(\frac{1}{2}\right)^2 = \sin^2\left(\frac{\pi}{6}\right)$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{6}, n \in I$$

$$\text{When } a = 8 \Rightarrow 16^{\sin^2 x} = 8$$

$$\Rightarrow 2^{4\sin^2 x} = 2^3$$

$$\Rightarrow 4\sin^2 x = 3$$

$$\Rightarrow \sin^2 x = \left(\frac{\sqrt{3}}{2}\right)^2 = \sin^2\left(\frac{\pi}{3}\right)$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{3}, n \in I$$

$$\text{Hence the solution set is } \left\{n\pi \pm \frac{\pi}{3}, n\pi \pm \frac{\pi}{3}\right\}, n \in I.$$

255. We have,

$$2^x + 2^{x-1} + 2^{x-2} = 5^x + 5^{x-1} + 5^{x-2}$$

$$\Rightarrow 2^x \left(1 + \frac{1}{2} + \frac{1}{2^2}\right) = 5^x \left(1 + \frac{1}{5} + \frac{1}{5^2}\right)$$

$$\Rightarrow 2^x \left(\frac{4+2+1}{4}\right) = 5^x \left(\frac{25+5+1}{25}\right)$$

$$\Rightarrow 2^x \left(\frac{7}{4}\right) = 5^x \left(\frac{31}{25}\right)$$

$$\Rightarrow \left(\frac{2}{5}\right)^x = \left(\frac{124}{175}\right)$$

$$\Rightarrow x = \log_2 \left(\frac{124}{175}\right)$$

$$\text{Hence the solution is } x = \log_2 \left(\frac{124}{175}\right).$$

256. We have,

$$(5+2\sqrt{6})^{x^2-3} + (5-2\sqrt{6})^{x^2-3} = 10.$$

$$\Rightarrow (5+2\sqrt{6})^{x^2-3} + \left(\frac{1}{5+2\sqrt{6}}\right)^{x^2-3} = 10$$

$$\Rightarrow (5+2\sqrt{6})^{x^2-3} + \frac{1}{(5+2\sqrt{6})^{x^2-3}} = 10$$

$$\Rightarrow a + \frac{1}{a} = 10, a = (5+2\sqrt{6})^{x^2-3}$$

$$\Rightarrow a^2 - 10a + 1 = 0$$

$$\Rightarrow a = \frac{10 \pm \sqrt{100-4}}{2}$$

$$= \frac{10 \pm 4\sqrt{6}}{2} = 5 \pm 2\sqrt{6}$$

$$\text{When } a = 5 + 2\sqrt{6}$$

$$\Rightarrow (5+2\sqrt{6})^{x^2-3} = (5+2\sqrt{6})$$

$$\Rightarrow x^2 - 3 = 1$$

$$\Rightarrow x^2 = 4$$

$$\Rightarrow x = \pm 2$$

$$\text{When } a = 5 - 2\sqrt{6}$$

$$\Rightarrow (5+2\sqrt{6})^{x^2-3} = (5-2\sqrt{6})$$

$$\Rightarrow (5+2\sqrt{6})^{x^2-3} = (5+2\sqrt{6})^{-1}$$

$$\Rightarrow x^2 - 3 = -1$$

$$\Rightarrow x^2 = 2$$

$$\Rightarrow x = \pm\sqrt{2}$$

$$\text{Hence the solution set is } \{-2, -\sqrt{2}, \sqrt{2}, 2\}$$

257. We have,

$$(15+4\sqrt{14})^x + (15-4\sqrt{14})^x = 30$$

$$\Rightarrow (15+4\sqrt{14})^x + \frac{1}{(15+4\sqrt{14})^x} = 30$$

$$\Rightarrow a + \frac{1}{a} = 30, \text{ where } a = 15 + 4\sqrt{14}$$

$$\text{Also, } a^2 - 30a + 1 = 0$$

$$\Rightarrow a = \frac{30 \pm \sqrt{900-4}}{2}$$

$$= \frac{30 \pm 8\sqrt{14}}{2} = 15 \pm 4\sqrt{14}$$

$$\text{When } a = 15 + 4\sqrt{14}$$

$$\Rightarrow (15+4\sqrt{14})^x = (15+4\sqrt{14})$$

$$\Rightarrow x = 1$$

$$\text{When } a = (15 - 4\sqrt{14})$$

$$\Rightarrow (15+4\sqrt{14})^x = (15-4\sqrt{14})$$

$$\Rightarrow (15+4\sqrt{14})^x = (15+4\sqrt{14})^{-1}$$

$$\Rightarrow x = -1$$

$$\text{Hence the solution set is } \{-1, 1\}.$$

258. We have $(5^x + 5^{-x}) = \log_{10} 25, x \in R$

The value of $5^x + 5^{-x} \geq 2$, where as the RHS < 2 .

Hence, the given equation has no solution.

259. The value of the LHS is $[-1, 1]$, whereas the

$$\text{RHS} = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$$

Graphical approach.

Hence the given equation has no solution.

260. We have,

$$p^x + p^{-x} = \log_q q^2 - 1$$

$$p^x + p^{-x} = \log_q q^2 - 1 = 2 - 1 = 1 = 2 - 1 = 1$$

The value of the LHS is ≥ 2 , whereas the RHS is 1

Hence the given equation has no solution.

261. We have,

$$n^{-|x|} |m - x| = 1$$

$$\Rightarrow |m - |x|| = n^{|x|}$$

$$\Rightarrow ||x| - m| = n^{|x|}$$

Graphical approach:

Hence the number of solutions is 1.

262. Given equation is $2x + x^2 = 1$

$$2x = 1 - x^2$$

Graphical approach.

It is satisfies for $x = 0$

Hence the number of solution is 1.

263. We have,

$$\begin{aligned} 2^x &> 1 \\ \Rightarrow 2^x &> 2^0 \\ \Rightarrow x &> 0 \end{aligned}$$

Hence the solution set is $(0, \infty)$.

264. We have,

$$\begin{aligned} \left(\frac{1}{3}\right)^x &> 1 \\ \Rightarrow \left(\frac{1}{3}\right)^x &> \left(\frac{1}{3}\right)^0 \\ \Rightarrow x &< 0 \end{aligned}$$

Hence the solution set is $(-\infty, 0)$.

265. We have,

$$\begin{aligned} 5^{x^2-3x+3} &> 5 \\ \Rightarrow 5^{x^2-3x+3} &> 5^1 \\ \Rightarrow x^2-3x+3 &> 1 \\ \Rightarrow x^2-3x+2 &> 0 \\ \Rightarrow (x-1)(x-2) &> 0 \\ \Rightarrow x &< 1 \text{ and } x > 2 \end{aligned}$$

Hence the solution set is $(-\infty, 1) \cup (2, \infty)$.

266. We have,

$$\begin{aligned} \left(\frac{1}{2}\right)^{x^2-5x+8} &> \left(\frac{1}{4}\right) \\ \Rightarrow \left(\frac{1}{2}\right)^{x^2-5x+8} &> \left(\frac{1}{2}\right)^2 \\ \Rightarrow x^2-5x+8 &> 2 \\ \Rightarrow x^2-5x+6 &> 0 \\ \Rightarrow (x-2)(x-3) &> 0 \\ \Rightarrow x &< 2 \text{ and } x > 3 \end{aligned}$$

Hence the solution set is $(-\infty, 2) \cup (3, \infty)$.

267. We have,

$$\begin{aligned} 4^x + 2^{x+1} - 8 &\leq 0 \\ \Rightarrow (2x)^2 + 2.2^x - 8 &\leq 0 \\ \Rightarrow a^2 + 2a - 8 &\leq 0 \\ \Rightarrow (a+4)(a-2) &\leq 0 \\ \Rightarrow -4 &\leq a \leq 2 \\ \Rightarrow -4 &\leq 2^x \leq 2 \\ \Rightarrow 0 < 2^x &\leq 2 \\ \Rightarrow -\infty < x &\leq 1 \end{aligned}$$

Hence the solution set is $(-\infty, 1)$.

268. We have,

$$\begin{aligned} |x|^{x^2-x-2} &\geq 1, -1 < x < 1 \\ \Rightarrow |x|^{x^2-x-2} &\geq x^0 \\ \Rightarrow x^2-x-2 &\leq 0 \\ \Rightarrow (x-2)(x+1) &\leq 0 \\ \Rightarrow -1 &\leq x \leq 2 \end{aligned}$$

Hence the solution set is $[-1, 2]$.

269. We have,

$$\begin{aligned} \left(\frac{1}{3}\right)^{\sqrt{x+4}} &> \left(\frac{1}{3}\right)^{\sqrt{x^2+3x+4}} \\ \Rightarrow \sqrt{x+4} &< \sqrt{x^2+3x+4} \end{aligned}$$

$$\begin{aligned} \Rightarrow x^2+3x+4 &> x+4, x \geq -4 \\ \Rightarrow x^2+2x &> 0, x \geq -4 \\ \Rightarrow x(x+2) &> 0, x \geq -4 \\ \Rightarrow x \leq -2, x \geq 0, x &\geq -4 \end{aligned}$$

Hence, the solution set is $[-4, 2] \cup [0, \infty)$.

270. We have,

$$\begin{aligned} \frac{2^{1-x} - 2^x + 1}{2^x - 1} &\leq 0 \\ \Rightarrow \frac{2 - (2^x)^2 + 2^x}{2^x(2^x - 1)} &\leq 0 \\ \Rightarrow \frac{(2^x)^2 - 2^x - 2}{2^x(2^x - 1)} &\geq 0 \\ \Rightarrow \frac{(2^x - 2)(2^x + 1)}{2^x(2^x - 1)} &\geq 0 \\ \Rightarrow \frac{(2^x - 2)}{(2^x - 1)} &\geq 0 \\ \Rightarrow 2^x \geq 2, 2^x < 1 \\ \Rightarrow x \geq 1, x < 0 \end{aligned}$$

Hence the solution set is $(-\infty, 0) \cup [1, \infty)$.

271. We have,

$$\begin{aligned} 2^{x+2} - 2^{x+3} - 2^{x+4} &> 5^{x+1} - 5^{x+2} \\ \Rightarrow 2^{x+2}(1 - 2 - 2^2) &> 5^{x+2}(5^{-1} - 1) \\ \Rightarrow 2^{x+2} \times (-5) &> 5^{x+2} \left(-\frac{4}{5}\right) \\ \Rightarrow \frac{2^{x+2}}{5^{x+2}} &< \frac{4}{25} \\ \Rightarrow \left(\frac{2}{5}\right)^{x+2} &< \left(\frac{2}{5}\right)^2 \\ \Rightarrow x+2 &> 2 \\ \Rightarrow x &> 0 \end{aligned}$$

Hence the solution set is $(0, \infty)$.

272. We have,

$$\begin{aligned} 3^{x^2-2} &< 2^{x^2-x} \\ \Rightarrow \frac{3^{x^2-x}}{2^{x^2-x}} &< 1 \\ \Rightarrow \left(\frac{3}{2}\right)^{x^2-x} &< \left(\frac{3}{2}\right)^0 \\ \Rightarrow x^2-x &< 0 \\ \Rightarrow x(x-1) &< 0 \\ \Rightarrow 0 < x &< 1 \end{aligned}$$

Hence the solution set is $(0, 1)$.